

Covariance-Scale Calibration of Empirical Populations: Theory and Applications to Resampling and Machine Learning

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Abstract

The empirical distribution assigns mass $1/n$ to each observation, so its covariance uses denominator n and has expectation $(n-1)\Sigma/n$ under i.i.d. sampling. We characterize a deterministic data-layer realization of the corresponding Bessel covariance-scale adjustment. Given observations $x_1, \dots, x_n \in \mathbb{R}^p$, define

$$y_i = \bar{x} + \sqrt{\frac{n}{n-1}}(x_i - \bar{x}), \quad i = 1, \dots, n. \quad (1)$$

The transformed empirical support preserves the sample mean and has denominator- n covariance exactly equal to the usual denominator- $(n-1)$ sample covariance of the original data. Under coordinate-wise affine separability, strict marginal monotonicity, and positive empirical marginal variances, this map is unique; under fixed observation correspondence, it uniquely minimizes total squared Euclidean displacement subject to the prescribed mean and covariance. For smooth fixed-population first-order projections of fixed-degree U -statistics, the induced perturbation is first-order asymptotically negligible. Under an explicitly assumed conditional bootstrap-validity statement, the surrogate empirical measure satisfies the required measure-level premises. The transformation requires $O(np)$ time. Simulations and a gene-expression illustration distinguish exact covariance-scale identities from algorithm-dependent downstream effects. In particular, covariance-derived functionals with known homogeneity may be recovered by analytical post-hoc rescaling, whereas the transformation does not itself guarantee improved estimation, prediction, clustering, or interval coverage.

Keywords: covariance-scale calibration; empirical measure; bootstrap resampling; machine learning; affine invariance

1 Introduction

Resampling methods—ranging from the classical bootstrap to modern ensemble learning techniques like bagging and subsampling—serve as the foundation for uncertainty quantification and predictive stabilization in machine learning. At the core of these methods is the plug-in principle: the observed sample $X = \{x_1, \dots, x_n\}$ is treated as the empirical distribution $\hat{P}_n$, which acts as a surrogate for the unknown true data-generating distribution P . Algorithms subsequently draw from $\hat{P}_n$ to simulate the sampling variability of the underlying universe.

The bootstrap (Efron, 1979) has become one of the most influential tools in modern statistics for uncertainty quantification, variance estimation, and predictive inference. Its theoretical foundations were subsequently established under broad regularity conditions by Bickel and Freedman (1981), while comprehensive treatments are given by Efron and Tibshirani (1993) and Davison and Hinkley (1997). Bootstrap methodology now underlies numerous machine learning procedures including ensemble learning, uncertainty estimation, and model aggregation.

The object of study is a data-layer transformation of the empirical support used by bootstrap, bagging, subsampling, and related learning pipelines. Rather than merely changing the denominator in a covariance estimator, the transformation changes the finite empirical measure supplied to a compatible downstream procedure. The paper characterizes when this replacement has an exact algebraic consequence, when it is equivalent to post-hoc rescaling, and when it can be irrelevant or alter a scale-sensitive pipeline.

The contribution is therefore not a new covariance estimator and not a claim that Bessel’s correction universally improves resampling. The central question is not whether denominator- n covariance is generally inferior to denominator- $(n - 1)$ covariance. Rather, it is when a prescribed finite-sample covariance-scale convention can be represented at the empirical-support level and how that representation propagates through a specified resampling or learning procedure. It is a characterization of a support-level intervention in empirical-measure-based learning systems: we identify its unique constrained affine realization, its fixed-correspondence minimum-displacement property, its exact propagation through homogeneous point-cloud functionals, and the conditions under which it is either equivalent to post-hoc rescaling or can change a scale-sensitive downstream pipeline.

The motivation is particularly relevant in modern machine learning, where bootstrap resampling forms the basis of bagging and numerous ensemble methods. Since predictive uncertainty directly depends on resampling variability, global covariance-scale calibration can affect the dispersion of resampling distributions and associated uncertainty summaries (Breiman, 1996).

1.1 The Empirical Covariance-Scale Convention and Data-Layer Calibration

The empirical measure $\hat{P}_n$ is the canonical nonparametric estimator of an unknown distribution P . Its covariance, defined with the empirical denominator n , is

$$\hat{\Sigma}_{\text{emp}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top.$$

Under i.i.d. sampling from a distribution with covariance Σ ,

$$\mathbb{E}[\widehat{\Sigma}_{\text{emp}}] = \frac{n-1}{n}\Sigma.$$

By contrast, the usual sample covariance with denominator $n-1$ is unbiased for Σ . These are distinct finite-sample objects: the former is the covariance of the empirical distribution, whereas the latter is an unbiased estimator of the population covariance. The factor $(n-1)/n$ is an expectation-level normalization relation; it does not assert a pointwise Loewner, spectral, entrywise, or Frobenius-norm ordering between a realized empirical covariance matrix and Σ .

This paper studies a deterministic data-layer representation of the denominator- $(n-1)$ covariance target. Specifically, it expands the empirical support about its sample mean so that the denominator- n covariance of the transformed support equals the usual denominator- $(n-1)$ sample covariance of the original observations. We refer to this replacement of the denominator- n empirical covariance scale by the denominator- $(n-1)$ sample-covariance scale as *covariance-scale calibration*.

What is new. The novelty of the present construction is not Bessel’s correction itself, which is classical, but its representation as a deterministic affine transformation of the empirical support. This data-layer representation realizes the denominator- $(n-1)$ covariance scale before a compatible resampling or learning pipeline is applied. It yields three consequences that are not supplied merely by changing the denominator in a covariance estimator: (i) under the stated coordinate-wise affine and monotonicity constraints, the transformation is uniquely characterized and, under fixed observation correspondence, is the minimum-displacement realization of the prescribed mean and covariance constraints; (ii) it gives exact propagation rules for point-cloud functionals and for downstream procedures whose response to common affine scaling is known; and (iii) it permits direct analysis of the perturbation of the empirical support and its first-order asymptotic consequences under stated regularity conditions. The framework therefore separates the exact algebraic covariance-scale identity from the algorithm-specific response of downstream resampling and learning procedures.

The resulting transformation is not a universal correction to bootstrap or machine-learning procedures. Whether replacing the empirical covariance scale improves a particular estimator, predictive procedure, or uncertainty summary depends on the statistical functional, tuning rule, loss, and data domain. In particular, procedures that are invariant to common positive affine rescaling may be unaffected, whereas scale-sensitive procedures can change when their hyperparameters are held fixed.

Several existing approaches address finite-sample covariance normalization at the estimator level through studentization, bias-corrected bootstrap methods, jackknife corrections, and other post-hoc adjustments. Such methods can be effective for specific estimators, but may be difficult to derive, analyze, or implement when the downstream object is a complex learning pipeline.

Our approach instead modifies the empirical support before resampling or model fitting. We construct a deterministic *Surrogate Population* $\mathbf{Y}$ that preserves the sample mean of the original data $\mathbf{X}$ and whose denominator- n empirical covariance equals the usual denominator- $(n-1)$ sample covariance $\mathbf{S}$ of $\mathbf{X}$. Subject to feature-domain compatibility and

any required handling of scale-dependent hyperparameters, existing bootstrap and learning pipelines can then be run on $\mathbf{Y}$ without changing their internal algorithmic steps.

The remainder of the paper develops this empirical-support representation. Section 2 establishes its finite-sample geometric and optimality properties, Sections 2.5–2.7 give first-order projection and conditional bootstrap-premise results, Sections 3–4 characterize algorithmic and computational implications, and Section 5 maps the resulting finite-sample consequences empirically.

1.2 Scope of the Data-Layer Construction

The surrogate population supplies an exact finite-sample covariance-scale identity. It does not by itself imply improved estimation error, predictive performance, or interval coverage for every downstream procedure. Those consequences depend on the statistical functional, learning algorithm, tuning rule, loss function, and sensitivity to global affine rescaling. The intended scope is therefore continuous, variance-sensitive settings in which the transformation respects the relevant data representation and inferential target.

Contributions

Our contributions are fourfold:

1. **Exact finite-sample covariance calibration.** We introduce the *Surrogate Population*, a sample-dependent deterministic affine transformation that preserves the sample mean while producing a transformed dataset whose empirical covariance, computed with denominator n , is exactly equal to the denominator- $(n - 1)$ sample covariance of the original data.
2. **Uniqueness under coordinate-wise affine constraints and minimal-distortion optimality.** Among affine maps that preserve the sample mean, coordinate-wise separability, and strict marginal monotonicity, and subject to positive empirical marginal variances, the surrogate transformation is unique. Separately, among all point configurations with the prescribed mean and covariance under fixed observation correspondence, it uniquely minimizes total squared Euclidean displacement.
3. **First-order projection equivalence and conditional bootstrap transfer.** For smooth population first-order projections, we establish asymptotic equivalence between evaluations on the original and surrogate samples. Under an explicitly assumed conditional bootstrap-validity property, we show that the surrogate empirical measure satisfies the measure-level premises of the assumed conditional bootstrap-validity property.
4. **Computational implementation and empirical scope assessment.**

The proposed method requires only an $O(np)$ preprocessing step. It can be inserted before a downstream resampling or learning pipeline without modifying that pipeline’s internal code, provided that the transformed representation remains admissible, all required covariates are transformed consistently, and scale-dependent tuning parameters are handled appropriately. Across covariance estimation, bootstrap PCA, bagged Ridge

regression, bagged K -nearest neighbors (K -NN), and bootstrap K -means clustering, we demonstrate both the strengths and limitations of surrogate calibration for representative variance-sensitive machine learning tasks. A gene-expression illustration verifies the same covariance-scale and PCA-eigenvalue identities on an observed continuous data set, without claiming population-level inferential improvement.

Variance-sensitive procedures. Throughout the paper, we use the term *variance-sensitive* for a statistical functional or learning procedure whose output, objective, regularization, distance calculation, spectral quantity, or uncertainty assessment can change under a common positive affine rescaling of the feature space. This terminology is broader than covariance estimation alone.

1.3 Relation to Existing Approaches

The proposed construction is complementary to, rather than a replacement for, existing covariance estimation, resampling, and uncertainty-quantification methods. Its distinguishing feature is the level at which it operates: it deterministically changes the empirical support before a compatible downstream procedure is run. By contrast, Bessel correction changes the normalization of a covariance calculation, shrinkage modifies a covariance estimator under a specified estimation criterion, and studentization or bias-corrected bootstrap methods modify an estimator or interval construction.

Table 1 summarizes the distinction between estimator-level and empirical-support-level covariance-scale identities.

Standardization rescales coordinates separately and changes the covariance matrix through feature-specific scaling, whereas whitening applies a generally non-isotropic transformation that removes covariance structure. Ledoit–Wolf and OAS shrink a covariance estimator. The surrogate map does none of these: it applies one positive isotropic rescaling about the sample mean so that the denominator- n covariance of the transformed support equals the denominator- $(n - 1)$ sample covariance of the original sample.

For covariance-derived quantities with known homogeneity, surrogate preprocessing can be exactly equivalent to analytical post-hoc rescaling. Its distinct feature is that the denominator- $(n - 1)$ covariance scale is represented directly in the empirical support before a compatible downstream pipeline is run, rather than only in an adjusted estimator computed after the pipeline’s input representation has been fixed. This does not imply universal improvement in estimation, prediction, or coverage.

Feature standardization and whitening are also data-level transformations, but they target different geometric properties. Standardization rescales individual coordinates, while whitening applies a generally non-isotropic map that alters correlations and principal directions. The surrogate map instead applies one common positive expansion about the empirical mean, preserving centered distance ratios, angles, and principal directions while changing only the global covariance scale.

Conformal prediction addresses a distinct inferential objective: predictive coverage under exchangeability for a specified conformal construction; see, for example, Lei et al. (2018). It is therefore not a substitute for global covariance-scale calibration, nor does covariance-scale calibration replace conformal inference. A split-conformal reference analysis is reported in Online Appendix 2. Because the conformal and bootstrap procedures use different training–

Table 1: Conceptual comparison of approaches related to empirical covariance-scale calibration. The final column distinguishes an exact finite-sample covariance-scale identity for an adjusted estimator from one for the covariance of a transformed empirical support. Neither type of identity implies universal improvement in downstream estimation or inference.

Approach	Primary target	Modifies data?	Applied before pipeline?	Exact finite-sample covariance-scale identity
Bessel correction	Covariance-estimator normalization	No	No	Yes, estimator-level
Studentization / BCa	Estimator- or interval-specific inference	No	No	No
Covariance shrinkage	Covariance estimation under a specified matrix-loss criterion	No	No	No
Standardization / whitening	Feature representation and coordinate scale	Yes	Yes	No
Variance-reduced bootstrap	Monte Carlo error from finite resampling	No	No	No
Conformal prediction	Predictive coverage under exchangeability	No	No	No
Surrogate population	Denominator- n covariance scale of the empirical support	Yes	Yes	Yes, empirical-support level

Note: For Bessel correction, the identity concerns an adjusted covariance estimator computed from the original observations. For the surrogate population, the identity concerns the denominator- n covariance of the transformed empirical support itself.

calibration constructions and target different inferential objects, those results are not a direct comparison of competing statistical guarantees.

The methodological contribution is therefore not a universal correction for machine-learning uncertainty. Rather, it is a deterministic finite-sample covariance-scale calibration layer together with an analysis characterizing its geometric, first-order asymptotic, and algorithm-dependent consequences. The experiments are designed to identify both settings in which this calibration changes a downstream procedure and settings in which affine invariance makes the correction irrelevant. Accordingly, the method should be evaluated relative to the downstream objective, tuning rule, loss function, and feature-domain constraints rather than as a universally beneficial preprocessing step. The central empirical message is that covariance-scale calibration is exact at the empirical-population level, but its downstream statistical value depends on the response of the relevant functional, tuning rule, and loss to common positive affine rescaling. From a methodological perspective, this paper is best viewed as a study of how a classical finite-sample covariance normalization can be represented at the empirical-support level and then propagated, or shown not to propagate, through downstream resampling and learning procedures.

2 The Surrogate Population Framework

This section develops a deterministic empirical-support representation of the denominator- $(n - 1)$ covariance scale. Rather than modifying a covariance estimator after it is computed, the construction transforms the empirical support before a compatible resampling or learning procedure is applied. Let $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ denote a set of n independent and identically distributed observations in $\mathbb{R}^p$, drawn from a distribution P with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. Let $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ denote the sample mean. The standard empirical covariance matrix is given by $\mathbf{S}_{\text{emp}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$, whose expectation satisfies

$$\mathbb{E}[\mathbf{S}_{\text{emp}}] = \frac{n-1}{n} \boldsymbol{\Sigma}.$$

The classical unbiased estimator is $\mathbf{S} = \frac{n}{n-1} \mathbf{S}_{\text{emp}}$. Our objective is to construct a transformed empirical support

$$Y = \{y_1, \dots, y_n\},$$

which we term the *Surrogate Population*, such that its raw empirical covariance exactly matches the unbiased target $\mathbf{S}$.

The construction should not be confused with covariance shrinkage, feature standardization, or whitening. Shrinkage modifies a covariance estimator under a specified estimation criterion, whereas the surrogate map changes the empirical support before resampling or learning. Standardization and whitening alter coordinate-wise scales or covariance structure; by contrast, the surrogate map applies one common positive affine expansion about the sample mean and therefore preserves the point cloud’s relative Euclidean geometry. Table 1 summarizes these distinctions.

2.1 Definition of the Surrogate Population

We define the surrogate population via a deterministic, isotropic expansion of the original data points around their empirical center of mass.

Definition 1 (Surrogate Population) *Given a sample $\mathbf{X}$, the surrogate population $\mathbf{Y}$ is generated by applying the transformation:*

$$\mathbf{y}_i = s(\mathbf{x}_i - \bar{\mathbf{x}}) + \bar{\mathbf{x}}, \quad \text{for } i = 1, \dots, n \quad (2)$$

where the scaling factor is defined as $s = \sqrt{\frac{n}{n-1}}$.

Here, the term “population” refers to the finite empirical support from which the downstream resampling procedure draws; it does not denote the unknown data-generating population P .

Conditional determinism. Throughout, “deterministic” means deterministic conditional on the observed sample. Once $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is given, its sample mean and the factor

$$s = \sqrt{\frac{n}{n-1}}$$

are fixed, and the surrogate population $\mathbf{Y}$ is uniquely determined by the stated map. The construction introduces no additional randomization; its randomness before conditioning arises only through the observed sample itself.

By construction, this transformation applies a common positive expansion of the empirical point cloud about its sample mean. A direct algebraic evaluation shows that the surrogate population exactly preserves the original sample mean ($\bar{\mathbf{y}} = \bar{\mathbf{x}}$), while its empirical covariance satisfies:

$$\frac{1}{n} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top = s^2 \mathbf{S}_{\text{emp}} = \frac{n}{n-1} \mathbf{S}_{\text{emp}} = \mathbf{S} \quad (3)$$

Geometric preview. The surrogate transformation changes the absolute centered scale of the empirical support while preserving its relative Euclidean geometry. Indeed, for every pair of observations,

$$\|\mathbf{y}_i - \mathbf{y}_j\|_2 = s \|\mathbf{x}_i - \mathbf{x}_j\|_2, \quad s = \sqrt{\frac{n}{n-1}}.$$

Thus all pairwise distances are multiplied by the same positive factor, while distance ratios, angles, and nearest-neighbor identities in the absence of distance ties are preserved. The formal point-cloud geometric consequences are given in Proposition 6, while the covariance and PCA scaling consequences are given in Corollary 7. Additional qualifications needed for algorithm-specific conclusions are developed in Section 3.

This identity concerns the empirical resampling population itself. The denominator- n covariance of $\mathbf{Y}$ equals the usual denominator- $(n-1)$ sample covariance of $\mathbf{X}$. It does not imply that an arbitrary estimator subsequently applied to surrogate data is unbiased or optimal under a general loss function.

Population calibration versus unbiased estimation. The identity

$$\mathbf{S}_Y = \frac{n}{n-1} \mathbf{S}_{\text{emp}} = \mathbf{S}$$

is an exact finite-sample calibration identity conditional on the observed sample. It does not imply that every downstream covariance estimator computed from $\mathbf{Y}$ is unbiased for the

population covariance Σ . Shrinkage, regularization, nonlinear estimation, and finite-sample optimization can introduce additional bias or loss that is not removed by the surrogate transformation. The target of the proposed preprocessing step is the global covariance scale of the empirical population, not universal unbiasedness of downstream covariance or precision estimators.

2.2 Affine Uniqueness

The definition above gives one transformation satisfying the desired covariance-scale identity. If covariance calibration alone were the objective, many alternative data perturbations, including the addition of independent noise, could produce a specified second moment. We therefore impose additional structural conditions: preservation of the sample mean, coordinate-wise separability, and strict marginal-order preservation. Under these stated affine constraints, the surrogate transformation is mathematically unique.

Coordinate-wise separability and marginal-order preservation. For the surrogate transformation to preserve the interpretation and ordering of individual features, we impose a stronger condition than preservation of the usual componentwise partial order. Namely, for each output coordinate j , the value $f_j(\mathbf{x})$ may depend only on the corresponding input coordinate x_j and must be strictly increasing in that coordinate. Equivalently, if two points have the same j th coordinate, then their transformed j th coordinates must also be equal (i.e., $f_j(\mathbf{x})$ is a function of x_j alone), regardless of their values on the remaining coordinates. For an affine map $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$, this condition forces $A_{j\ell} = 0$ for every $\ell \neq j$, since otherwise the j th output coordinate would vary when only x_ℓ changes. Strict monotonicity in x_j further requires $A_{jj} > 0$. Hence, $\mathbf{A}$ must be a positive diagonal matrix. This coordinate-wise separability condition is stronger than ordinary preservation of the componentwise partial order, which by itself would allow nonnegative off-diagonal entries.

Theorem 2 (Sample-conditional uniqueness under coordinate-wise affine calibration)

Let $f : \mathbb{R}^p \rightarrow \mathbb{R}^p$ be an affine transformation $\mathbf{y}_i = \mathbf{A}\mathbf{x}_i + \mathbf{b}$. Suppose f satisfies the following conditions:

1. **Mean preservation:** $\frac{1}{n} \sum_{i=1}^n f(\mathbf{x}_i) = \bar{\mathbf{x}}$.
2. **Exact covariance calibration:** The empirical covariance of $\{f(\mathbf{x}_i)\}_{i=1}^n$ equals the unbiased sample covariance $\mathbf{S}$.
3. **Coordinate-wise separability and marginal-order preservation:** For every coordinate j , $f_j(\mathbf{x})$ depends only on x_j and is strictly increasing in x_j (equivalently, for an affine map, $\mathbf{A}$ is a positive diagonal matrix).
4. **Positive marginal variance:** $(\mathbf{S}_{\text{emp}})_{jj} > 0$ for each $j = 1, \dots, p$.

Then, for the observed sample $\{\mathbf{x}_i\}_{i=1}^n$ satisfying the stated positive-marginal-variance condition, f is unique within the specified coordinate-wise affine class and is given by $\mathbf{A} = \sqrt{\frac{n}{n-1}} \mathbf{I}_p$ and $\mathbf{b} = \left(1 - \sqrt{\frac{n}{n-1}}\right) \bar{\mathbf{x}}$, which is identically the surrogate transformation.

Scope of uniqueness. Theorem 2 establishes uniqueness only within the stated coordinate-wise affine class. If coordinate-wise separability and strict marginal monotonicity are removed, additional affine maps can satisfy the mean and covariance constraints, depending on the geometry and rank structure of S_{emp} . Thus the structural restrictions are part of the characterization, not merely technical conditions.

Role of the positive-variance condition. The uniqueness conclusion is conditional on the observed sample satisfying

$$(S_{\text{emp}})_{jj} > 0, \quad j = 1, \dots, p.$$

If $(S_{\text{emp}})_{jj} = 0$ for some coordinate j , then $x_{ij} = \bar{x}_j$ for every i , and the covariance constraint does not identify the corresponding positive diagonal scaling coefficient a_j . Mean preservation then sets

$$b_j = (1 - a_j)\bar{x}_j,$$

so every $a_j > 0$ yields the same transformed observed values in that coordinate. Thus the affine map is not unique on degenerate coordinates, although the transformed observed values in those coordinates remain identical for every $a_j > 0$. Equivalently, uniqueness holds on the non-degenerate coordinate subspace.

Sample-conditional nature of the condition. The condition $(S_{\text{emp}})_{jj} > 0$ is a sample-conditional identification condition rather than a population assumption. If $\text{Var}_P(X_j) > 0$, then under i.i.d. non-degenerate sampling

$$\mathbb{P}\{(S_{\text{emp}})_{jj} > 0\} \rightarrow 1.$$

The finite-sample uniqueness result itself, however, is conditional on the observed empirical variance being positive; it does not require a population variance assumption.

Proof Sketch. Condition 2 requires that $\mathbf{A}S_{\text{emp}}\mathbf{A}^\top = \mathbf{S} = \frac{n}{n-1}S_{\text{emp}}$. By Condition 3, coordinate-wise separability and strict marginal monotonicity require $\mathbf{A} = \text{diag}(a_1, \dots, a_p)$ with $a_j > 0$. Considering the diagonal entries of the covariance identity yields $a_j^2(S_{\text{emp}})_{jj} = \frac{n}{n-1}(S_{\text{emp}})_{jj}$. For positive empirical marginal variances, this gives

$$a_j = \sqrt{\frac{n}{n-1}}$$

for every j , and hence $\mathbf{A} = s\mathbf{I}_p$. Finally, mean preservation gives

$$\bar{\mathbf{y}} = \mathbf{A}\bar{\mathbf{x}} + \mathbf{b} = \bar{\mathbf{x}}.$$

Since $\mathbf{A} = s\mathbf{I}_p$, it follows that

$$\mathbf{b} = \bar{\mathbf{x}} - \mathbf{A}\bar{\mathbf{x}} = (\mathbf{I}_p - s\mathbf{I}_p)\bar{\mathbf{x}} = (1 - s)\bar{\mathbf{x}}.$$

Therefore,

$$\mathbf{y}_i = s\mathbf{x}_i + (1 - s)\bar{\mathbf{x}} = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}),$$

which is the stated surrogate transformation. The complete argument, including the degenerate-coordinate case, is given in Online Appendix 1.

2.3 Optimality via Minimal Distortion

Theorem 2 establishes uniqueness within the class of coordinate-wise affine transformations. We now allow arbitrary point configurations and ask which configuration satisfying the prescribed mean and covariance constraints minimizes pointwise displacement from the observed sample. The resulting optimization uses the fixed correspondence $\mathbf{x}_i \mapsto \mathbf{y}_i$ for $i = 1, \dots, n$.

Remark 3 (Fixed-correspondence interpretation) *Theorem 4 below is a finite-dimensional, fixed-correspondence optimization result. It does not solve an optimal-transport problem over arbitrary couplings, permutations, or empirical-measure matchings. Its objective is the total squared displacement under the prescribed correspondence $x_i \mapsto y_i$, $i = 1, \dots, n$.*

Theorem 4 (Minimal-Distortion Optimality) *Let $n > 1$ and let $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$ have sample mean $\bar{\mathbf{x}}$ and denominator- n empirical covariance $\mathbf{S}_{\text{emp}}$. Set*

$$\mathbf{S} = \frac{n}{n-1} \mathbf{S}_{\text{emp}}, \quad s = \sqrt{\frac{n}{n-1}}.$$

Consider point configurations $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_n)$ satisfying

$$\frac{1}{n} \sum_{i=1}^n \mathbf{y}_i = \bar{\mathbf{x}}$$

and

$$\frac{1}{n} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{x}})(\mathbf{y}_i - \bar{\mathbf{x}})^\top = \mathbf{S},$$

with the observation correspondence fixed by $\mathbf{x}_i \mapsto \mathbf{y}_i$. Within this fixed-correspondence feasible set, the surrogate population

$$\mathbf{y}_i = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}), \quad i = 1, \dots, n,$$

is the unique minimizer of

$$\sum_{i=1}^n \|\mathbf{y}_i - \mathbf{x}_i\|_2^2.$$

Proof sketch. Let $\mathbf{X}_c$ and $\mathbf{Y}_c$ denote the centered row matrices associated with the fixed correspondence $\mathbf{x}_i \mapsto \mathbf{y}_i$, $i = 1, \dots, n$. That is,

$$X_c = \begin{bmatrix} (x_1 - \bar{x})^\top \\ \vdots \\ (x_n - \bar{x})^\top \end{bmatrix}, \quad Y_c = \begin{bmatrix} (y_1 - \bar{x})^\top \\ \vdots \\ (y_n - \bar{x})^\top \end{bmatrix}.$$

The covariance constraint is equivalent to

$$Y_c^\top Y_c = s^2 X_c^\top X_c, \quad s = \sqrt{\frac{n}{n-1}},$$

so $Y_c = sX_c$ is feasible. Moreover, every feasible Y_c satisfies

$$\|Y_c\|_F^2 = \text{tr}(Y_c^\top Y_c) = s^2 \text{tr}(X_c^\top X_c) = s^2 \|X_c\|_F^2.$$

Hence,

$$\begin{aligned} \|\mathbf{Y}_c - \mathbf{X}_c\|_F^2 &= \|\mathbf{Y}_c\|_F^2 + \|\mathbf{X}_c\|_F^2 - 2\langle \mathbf{Y}_c, \mathbf{X}_c \rangle_F \\ &\geq s^2 \|\mathbf{X}_c\|_F^2 + \|\mathbf{X}_c\|_F^2 - 2s \|\mathbf{X}_c\|_F^2 \\ &= (s-1)^2 \|\mathbf{X}_c\|_F^2, \end{aligned}$$

where the inequality follows from Frobenius Cauchy–Schwarz:

$$\langle \mathbf{Y}_c, \mathbf{X}_c \rangle_F \leq \|\mathbf{Y}_c\|_F \|\mathbf{X}_c\|_F = s \|\mathbf{X}_c\|_F^2.$$

If $\mathbf{X}_c = \mathbf{0}$, then $\mathbf{S}_{\text{emp}} = \mathbf{0}$, and the covariance constraint forces $\mathbf{Y}_c = \mathbf{0}$. Hence the unique feasible configuration is $\mathbf{Y} = \mathbf{X}$. We therefore assume below that $\|\mathbf{X}_c\|_F > 0$. Equality holds if and only if $\mathbf{Y}_c = c\mathbf{X}_c$ for some scalar c . The covariance constraint then gives $c^2 = s^2$, and equality in the displayed lower bound selects $c = s > 0$. Therefore, $\mathbf{Y}_c = s\mathbf{X}_c$ is the unique feasible minimizer. The complete rank-free argument is given in Online Appendix 1.

Proposition 5 (Affine propagation of point-cloud functionals) *Let*

$$\mathbf{y}_i = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}), \quad s > 0, \quad i = 1, \dots, n,$$

and define

$$\mathcal{A}_{\bar{\mathbf{x}},s}(\mathbf{z}) = \bar{\mathbf{x}} + s(\mathbf{z} - \bar{\mathbf{x}}) = (1-s)\bar{\mathbf{x}} + s\mathbf{z}, \quad \mathbf{z} \in \mathbb{R}^p.$$

Then the denominator- n empirical covariance satisfies

$$\mathbf{S}_{Y,\text{emp}} = s^2 \mathbf{S}_{X,\text{emp}}.$$

More generally, let T be a functional of a finite point cloud. If T is invariant under common translations and positive scalar rescaling, so that

$$T(\{\mathbf{a} + b\mathbf{x}_i\}_{i=1}^n) = T(\{\mathbf{x}_i\}_{i=1}^n)$$

for every $\mathbf{a} \in \mathbb{R}^p$ and $b > 0$, then

$$T(\mathbf{Y}) = T(\mathbf{X}).$$

If instead T is translation invariant and homogeneous of degree r under positive scalar rescaling, so that

$$T(\{\mathbf{a} + b\mathbf{x}_i\}_{i=1}^n) = b^r T(\{\mathbf{x}_i\}_{i=1}^n)$$

for every $\mathbf{a} \in \mathbb{R}^p$ and $b > 0$, then

$$T(\mathbf{Y}) = s^r T(\mathbf{X}).$$

Proof Because

$$\mathbf{y}_i - \bar{\mathbf{y}} = s(\mathbf{x}_i - \bar{\mathbf{x}}),$$

the covariance identity follows immediately. Moreover,

$$\mathbf{y}_i = (1-s)\bar{\mathbf{x}} + s\mathbf{x}_i$$

is a common translation plus positive scalar rescaling of every point in $\mathbf{X}$. The invariance and homogeneity conclusions therefore follow directly from the defining properties of T . ■

Proposition 6 (Global geometric invariance under surrogate calibration) *Let*

$$\mathbf{y}_i = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}), \quad s > 0, \quad i = 1, \dots, n.$$

Then, for every i, j ,

$$\mathbf{y}_i - \mathbf{y}_j = s(\mathbf{x}_i - \mathbf{x}_j), \quad \|\mathbf{y}_i - \mathbf{y}_j\|_2 = s\|\mathbf{x}_i - \mathbf{x}_j\|_2.$$

Consequently, all pairwise distances are multiplied by s , all nonzero pairwise-distance ratios are preserved, and for any nonzero difference vectors $\mathbf{x}_i - \mathbf{x}_j$ and $\mathbf{x}_k - \mathbf{x}_\ell$,

$$\frac{(\mathbf{y}_i - \mathbf{y}_j)^\top (\mathbf{y}_k - \mathbf{y}_\ell)}{\|\mathbf{y}_i - \mathbf{y}_j\|_2 \|\mathbf{y}_k - \mathbf{y}_\ell\|_2} = \frac{(\mathbf{x}_i - \mathbf{x}_j)^\top (\mathbf{x}_k - \mathbf{x}_\ell)}{\|\mathbf{x}_i - \mathbf{x}_j\|_2 \|\mathbf{x}_k - \mathbf{x}_\ell\|_2}.$$

Thus, the surrogate transformation preserves the Euclidean geometry of the observed point cloud up to a common positive scale factor.

Proof The identity

$$\mathbf{y}_i - \mathbf{y}_j = s(\mathbf{x}_i - \mathbf{x}_j)$$

follows directly from the definition of the surrogate transformation. The distance identity follows by taking Euclidean norms. Nonzero distance ratios and the displayed cosine identity follow by cancellation of the common positive factor s . $\blacksquare$

Corollary 7 (Exact covariance and PCA eigenvalue scaling) *Let*

$$S_{X,\text{emp}} = V\Lambda V^\top$$

be an eigendecomposition of the denominator- n empirical covariance matrix of the original sample X . Under surrogate calibration,

$$S_{Y,\text{emp}} = \frac{n}{n-1} S_{X,\text{emp}} = V \left(\frac{n}{n-1} \Lambda \right) V^\top.$$

Consequently, for every $j = 1, \dots, p$,

$$\lambda_j(S_{Y,\text{emp}}) = \frac{n}{n-1} \lambda_j(S_{X,\text{emp}}),$$

while the eigenspaces of $S_{X,\text{emp}}$ and $S_{Y,\text{emp}}$ coincide. In particular, an eigenvector associated with a simple eigenvalue may be chosen identically under both representations.

Proof Because

$$\mathbf{y}_i - \bar{\mathbf{y}} = s(\mathbf{x}_i - \bar{\mathbf{x}}), \quad s^2 = \frac{n}{n-1},$$

we have

$$S_{Y,\text{emp}} = s^2 S_{X,\text{emp}} = \frac{n}{n-1} S_{X,\text{emp}}.$$

Multiplication of a symmetric matrix by a positive scalar leaves its eigenspaces unchanged and multiplies every eigenvalue by that scalar. $\blacksquare$

Remark 8 (Scope of affine propagation) *Proposition 5 concerns functionals of the observed feature point cloud. It does not imply that an arbitrary learning algorithm is invariant or homogeneous under surrogate preprocessing. Such conclusions require additional conditions on the algorithm, its tuning parameters, and its inputs. For example, K -nearest-neighbor procedures require that both training and query covariates undergo the same training-derived transformation; under this condition, the corresponding distance-order preservation result is formalized in Corollary 12. Likewise, matched-initialization K -means requires correspondingly transformed initial centers, matched resamples when bootstrap behavior is considered, and consistent tie handling. Regularized procedures with fixed scale-dependent tuning parameters need not satisfy the homogeneity property, and bootstrap distributional equivalence does not follow from the point-cloud transformation alone.*

2.4 Scope of the Calibration Map

The surrogate construction is defined for observations

$$\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p, \quad n > 1.$$

Its exact finite-sample mean-preservation and covariance-scale identities are algebraic and require no distributional assumption. The uniqueness result in Theorem 2 additionally requires positive empirical variance in every transformed coordinate:

$$(\mathbf{S}_{\text{emp}})_{jj} > 0, \quad j = 1, \dots, p.$$

This condition is coordinatewise and does not require $\mathbf{S}_{\text{emp}}$ to be full rank. The asymptotic and conditional bootstrap-transfer results require the separate smoothness, moment, non-degeneracy where applicable, and bootstrap-stability assumptions stated in Sections 2.5–2.7.

The map is most naturally used for continuous feature representations in unconstrained Euclidean domains. Although it is algebraically defined for any real-valued input matrix, it can violate bounded-support, discreteness, integrality, positivity, compositional, or other domain constraints. Such settings require a separate support-preserving construction or a problem-specific justification.

Unless otherwise stated, the transformation is applied only to covariates supplied to the downstream learning or resampling procedure; response variables remain on their original scale. The transformation changes global centered covariance scale while preserving relative Euclidean geometry. Accordingly, its downstream consequences are algorithm-, tuning-rule-, loss-, and domain-dependent and are not universally beneficial.

Summary of exact and asymptotic claims. The surrogate construction has three logically distinct consequences. First, it satisfies the exact finite-sample identity

$$\mathbf{S}_{Y,\text{emp}} = \frac{n}{n-1} \mathbf{S}_{\text{emp}} = \mathbf{S},$$

where $\mathbf{S}_{\text{emp}}$ and $\mathbf{S}_{Y,\text{emp}}$ are the denominator- n empirical covariance matrices of the original and surrogate samples, respectively, and $\mathbf{S}$ is the usual denominator- $(n-1)$ sample covariance of the original observations. Second, under the structural restrictions in Section 2, the associated map is unique within the stated coordinate-wise affine class and is the unique

minimum-distortion point-cloud transformation satisfying the prescribed mean and covariance constraints. Third, because

$$s_n - 1 = \sqrt{\frac{n}{n-1}} - 1 = O(n^{-1}),$$

the induced average squared displacement is $O_P(n^{-2})$ under a finite second-moment condition. Consequently, for sufficiently smooth population first-order projections and related asymptotically linear functionals, the surrogate perturbation is first-order asymptotically negligible under the additional regularity conditions stated in Sections 2.5–2.7. These statements should be distinguished: covariance calibration is an exact finite-sample identity, whereas preservation of first-order inferential behavior is an asymptotic property requiring smoothness, moment, and conditional bootstrap-stability assumptions.

Section 2 establishes exact finite-sample global covariance-scale calibration. We now consider a fixed-dimensional asymptotic regime in which p is held fixed as $n \rightarrow \infty$, and examine smooth U -statistics and related asymptotically linear functionals of the surrogate population. Throughout this section, g_P denotes the population first-order projection associated with the fixed data-generating distribution P . We assume that the relevant projection is continuously differentiable with globally Lipschitz gradient, that

$$\mathbb{E}_P\{g_P^2(\mathbf{X})\} < \infty,$$

and that the additional moment conditions stated below hold. Under these smoothness and moment conditions, the fixed population first-order projections evaluated at the original and surrogate observations differ by $O_P(n^{-1/2})$. Non-degeneracy,

$$\text{Var}_P\{g_P(X)\} > 0,$$

is additionally required only for the stated first-order distributional consequence for the full U -statistic. When the relevant statistic admits a separately verified asymptotic-linear representation for both samples, this projection equivalence yields equality of their first-order asymptotic distributions. Combined with the explicitly assumed conditional bootstrap-validity condition in Theorem 11, the argument also yields a conditional premise-preservation result for the surrogate empirical measure.

Scope. The results are intended for variance-dependent, smooth functionals of continuous distributions. They do not extend to highly non-smooth statistics (indicator functionals, rank-based tests) or to degenerate U -statistics whose limiting distributions are governed by higher-order chaos. In those regimes, the $O_P(n^{-1})$ perturbation induced by the surrogate map need not transfer through the functional in a first-order manner, and classical nonparametric bootstrap procedures can fail or require statistic-specific modification for important non-smooth and degenerate functionals. The theory directly applies to smooth, non-degenerate fixed-degree U -statistics and can be extended to other functionals only when an appropriate asymptotic-linear expansion is verified separately. Covariance-based quantities provide direct examples. PCA eigenvalues additionally require the relevant eigenvalue-separation and differentiability conditions, while learner-specific prediction-error functionals require a separate analysis of fitting, tuning, and resampling effects.

2.5 Hájek Projection Equivalence and First-Order Asymptotics

Let $U_n(\mathbf{X})$ be a U -statistic of fixed degree m with symmetric kernel h , and let

$$\theta(P) = \mathbb{E}_P[h(\mathbf{X}_1, \dots, \mathbf{X}_m)].$$

For the centered statistic $U_n(\mathbf{X}) - \theta(P)$, define the fixed-population first-order Hájek projection term by

$$\hat{L}_n(\mathbf{X}) = \sum_{i=1}^n [\mathbb{E}_P\{U_n(\mathbf{X}) \mid \mathbf{X}_i\} - \theta(P)] = \frac{m}{n} \sum_{i=1}^n g_P(\mathbf{X}_i), \quad (3)$$

where

$$g_P(\mathbf{x}) = \mathbb{E}_P[h(\mathbf{x}, \mathbf{X}_2, \dots, \mathbf{X}_m)] - \theta(P).$$

Equivalently, the Hájek projection of the uncentered statistic $U_n(\mathbf{X})$ is

$$\theta(P) + \hat{L}_n(\mathbf{X}).$$

Here and below, g_P denotes the population first-order projection defined with respect to the fixed data-generating distribution P . Accordingly, $\hat{L}_n(\mathbf{Y})$ evaluates this same population projection function at the surrogate observations; it is not an empirical-measure-specific projection defined by replacing P with $\hat{P}_n^Y$.

A critical concern is whether applying the surrogate transformation $\mathbf{y}_i = s(\mathbf{x}_i - \bar{\mathbf{x}}) + \bar{\mathbf{x}}$ irreparably distorts the structural projection of the data. We demonstrate that for smooth kernels, the surrogate population preserves the Hájek projection up to asymptotically negligible terms.

Lemma 9 (Average Perturbation Induced by the Surrogate Transformation) *Assume $\mathbb{E}\|\mathbf{X}\|_2^2 < \infty$. Let*

$$\mathbf{y}_i = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}), \quad s = \sqrt{\frac{n}{n-1}}.$$

Then

$$\frac{1}{n} \sum_{i=1}^n \|\mathbf{y}_i - \mathbf{x}_i\|_2^2 = (s-1)^2 \text{tr}(\mathbf{S}_{\text{emp}}) = O_P(n^{-2}),$$

where

$$\mathbf{S}_{\text{emp}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$$

is the empirical covariance matrix. Consequently,

$$\left(\frac{1}{n} \sum_{i=1}^n \|\mathbf{y}_i - \mathbf{x}_i\|_2^2 \right)^{1/2} = O_P(n^{-1}),$$

so the surrogate transformation produces an asymptotically negligible root-mean-square perturbation of the observations.

Proof Since

$$\mathbf{y}_i - \mathbf{x}_i = (s - 1)(\mathbf{x}_i - \bar{\mathbf{x}}),$$

we have

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \|\mathbf{y}_i - \mathbf{x}_i\|_2^2 &= (s - 1)^2 \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^2 \\ &= (s - 1)^2 \text{tr}(\mathbf{S}_{\text{emp}}). \end{aligned}$$

Moreover,

$$s = \sqrt{\frac{n}{n-1}} = 1 + \frac{1}{2n} + O(n^{-2}),$$

and hence $(s-1)^2 = O(n^{-2})$. Under the finite-second-moment assumption, $\text{tr}(\mathbf{S}_{\text{emp}}) = O_P(1)$, which proves the first claim. Taking square roots yields the root-mean-square rate. $\blacksquare$

Define the fixed-population first-order projection evaluations by

$$\hat{L}_n(X) = \frac{m}{n} \sum_{i=1}^n g_P(x_i), \quad \hat{L}_n(Y) = \frac{m}{n} \sum_{i=1}^n g_P(y_i).$$

Theorem 10 (Asymptotic equivalence of population first-order projections) *Let $U_n(\mathbf{X})$ be a U -statistic of fixed degree m with symmetric kernel h and population first-order projection*

$$g_P(\mathbf{x}) = \mathbb{E}_P[h(\mathbf{x}, \mathbf{X}_2, \dots, \mathbf{X}_m)] - \theta(P),$$

where

$$\theta(P) = \mathbb{E}_P[h(\mathbf{X}_1, \dots, \mathbf{X}_m)].$$

Throughout this theorem, the ambient dimension p is fixed. Define

$$\hat{L}_n(\mathbf{X}) = \frac{m}{n} \sum_{i=1}^n g_P(\mathbf{X}_i), \quad \hat{L}_n(\mathbf{Y}) = \frac{m}{n} \sum_{i=1}^n g_P(\mathbf{Y}_i),$$

where

$$\mathbf{Y}_i = \bar{\mathbf{X}} + s_n(\mathbf{X}_i - \bar{\mathbf{X}}), \quad s_n = \sqrt{\frac{n}{n-1}}.$$

Suppose that g_P is continuously differentiable with globally Lipschitz gradient and that

$$\mathbb{E}_P \|\mathbf{X}\|_2^4 < \infty, \quad \mathbb{E}_P \|\nabla g_P(\mathbf{X})\|_2^2 < \infty.$$

Then

$$\sqrt{n} \left\{ \hat{L}_n(\mathbf{Y}) - \hat{L}_n(\mathbf{X}) \right\} \xrightarrow{P} 0.$$

Consequence for asymptotically linear statistics. *If, in addition, the statistic is non-degenerate,*

$$\text{Var}_P\{g_P(X)\} > 0,$$

and both statistics admit the asymptotic-linear representations

$$U_n(\mathbf{X}) - \theta(P) = \hat{L}_n(\mathbf{X}) + o_P(n^{-1/2})$$

and

$$U_n(\mathbf{Y}) - \theta(P) = \hat{L}_n(\mathbf{Y}) + o_P(n^{-1/2}),$$

then

$$\sqrt{n}\{U_n(\mathbf{Y}) - U_n(\mathbf{X})\} \xrightarrow{P} 0.$$

Thus, under the additional asymptotic-linear representation assumptions, $U_n(X)$ and $U_n(Y)$ have the same first-order asymptotic distribution.

This conclusion is for the fixed-dimensional regime specified in the assumptions. It does not establish an analogous first-order result when the ambient dimension grows with the sample size.

Two levels of conclusion. The first conclusion of Theorem 10 is a statement about the fixed population projection g_P evaluated at the original and surrogate point clouds. The second conclusion is a separate consequence that additionally requires asymptotic-linear representations for both $U_n(\mathbf{X})$ and $U_n(\mathbf{Y})$, centered at the same target $\theta(P)$. Projection equivalence alone does not imply

$$\sqrt{n}\{U_n(\mathbf{Y}) - U_n(\mathbf{X})\} \xrightarrow{P} 0;$$

that conclusion requires the additional asymptotic-linear representation assumptions stated in the theorem.

Proof sketch. Write $\Delta_i = \mathbf{y}_i - \mathbf{x}_i = (s_n - 1)(\mathbf{x}_i - \bar{\mathbf{x}})$. Lemma 9 gives $n^{-1} \sum_i \|\Delta_i\|_2^2 = \mathcal{O}_P(n^{-2})$. A first-order expansion of the fixed population projection g_P then yields

$$\hat{L}_n(\mathbf{Y}) - \hat{L}_n(\mathbf{X}) = \frac{m}{n} \sum_{i=1}^n \nabla g_P(\mathbf{x}_i)^\top \Delta_i + R_n,$$

where the linear term is $\mathcal{O}_P(n^{-1})$ by Cauchy–Schwarz and the remainder is $\mathcal{O}_P(n^{-2})$. Multiplying by $\sqrt{n}$ gives $o_P(1)$. The complete argument is given in Online Appendix 1.

2.6 Conditional Preservation of Bootstrap-Validity Premises

The surrogate transformation does not independently establish bootstrap consistency for arbitrary statistics or learning algorithms. Instead, under the conditions below, it preserves two measure-level premises—weak convergence to P and convergence of the relevant fixed projection moment—that occur in an explicitly assumed conditional bootstrap-validity statement.

Theorem 11 (Surrogate preservation of premises: assumed bootstrap-validity theorem)

Suppose that p is fixed, that g_P is continuously differentiable with globally Lipschitz gradient, and that

$$\mathbb{E}_P \|\mathbf{X}\|_2^4 < \infty, \quad \mathbb{E}_P \|\nabla g_P(\mathbf{X})\|_2^2 < \infty.$$

Assume additionally that

$$\mathbb{E}_P [g_P^2(\mathbf{X})] < \infty, \quad \mathbb{E}_P [\|\nabla g_P(\mathbf{X})\|_2^4] < \infty.$$

Assume the following conditional bootstrap-stability property: for every sequence of possibly random empirical probability measures Q_n satisfying

$$Q_n \Rightarrow P \quad \text{in probability,} \quad \int g_P^2 dQ_n \xrightarrow{P} \mathbb{E}_P [g_P^2(\mathbf{X})],$$

the conditional bootstrap distribution obtained by drawing n observations with replacement from Q_n satisfies

$$\sup_{t \in \mathbb{R}} |\mathbb{P}_{Q_n}^* \{ \sqrt{n}(U_n^* - U_n(Q_n)) \leq t \} - \mathbb{P} \{ \sqrt{n}(U_n(\mathbf{X}) - \theta(P)) \leq t \}| \xrightarrow{P} 0,$$

where $U_n(Q_n)$ denotes the observed statistic associated with the empirical measure Q_n .

Let $U_n^*(\mathbf{Y})$ denote the bootstrap statistic obtained by drawing n observations with replacement from the surrogate population $\mathbf{Y}$. Then

$$\sup_{t \in \mathbb{R}} |\mathbb{P}^* \{ \sqrt{n}(U_n^*(\mathbf{Y}) - U_n(\mathbf{Y})) \leq t \} - \mathbb{P} \{ \sqrt{n}(U_n(\mathbf{X}) - \theta(P)) \leq t \}| \xrightarrow{P} 0.$$

Here $U_n(Q_n)$ denotes the statistic computed from the observed support of Q_n , and U_n^* denotes its counterpart computed from a size- n resample drawn conditionally from Q_n . Throughout this result, the ambient dimension p is fixed.

The conclusion is conditional on the displayed bootstrap-stability assumption.

Consistency versus finite-sample calibration. Bootstrap consistency is an asymptotic distributional statement. It should not be interpreted as a guarantee of nominal finite-sample coverage for percentile intervals, prediction intervals, or other interval constructions. Likewise, the exact covariance-scale calibration established by the surrogate transformation concerns the empirical support and does not by itself calibrate a downstream inferential procedure. In particular, replacing the denominator- n covariance scale of the empirical support by the denominator- $(n-1)$ sample-covariance scale does not imply that intervals constructed from the surrogate population attain their nominal coverage at a finite sample size. This distinction is central to the empirical results in Section 5, where covariance-scale calibration can alter interval behavior without resolving the broader sources of estimation error or predictive uncertainty.

Logical status and assumptions. Theorem 11 is a conditional transfer result, not an independent bootstrap-consistency theorem. Its bootstrap-stability assumption is deliberately formulated at the level of the underlying statistic: it posits that a classical conditional bootstrap-validity theorem applies to empirical-measure sequences satisfying the displayed weak-convergence and projection-moment conditions. The contribution here is to verify that the surrogate empirical measure $\hat{P}_n^Y$ belongs to this admissible class:

$$\hat{P}_n^Y \Rightarrow P \quad \text{in probability,} \quad \int g_P^2 d\hat{P}_n^Y \xrightarrow{P} \mathbb{E}_P[g_P^2(\mathbf{X})].$$

The fourth-moment condition on $\nabla g_P(\mathbf{X})$ is used in Online Appendix 1 to transfer convergence of the empirical second moment of the fixed population projection score from $\hat{P}_n^X$ to $\hat{P}_n^Y$. Thus, bootstrap validity is not inferred from the affine transformation alone; whenever a statistic-specific bootstrap-validity theorem is available in the displayed form, its conclusion applies with $Q_n = \hat{P}_n^Y$.

Proof sketch. Pair each $\mathbf{x}_i$ with $\mathbf{y}_i$. Lemma 9 gives $W_2(\hat{P}_n^X, \hat{P}_n^Y) = \mathcal{O}_P(n^{-1})$. For fixed p and finite second moments, $W_2(\hat{P}_n^X, P) \rightarrow 0$ in probability, so the triangle inequality yields $\hat{P}_n^Y \Rightarrow P$. A Taylor expansion of the fixed score g_P , together with the fourth-moment bound on ∇g_P , transfers the empirical second-moment convergence from $\hat{P}_n^X$ to $\hat{P}_n^Y$. The assumed conditional bootstrap-stability property then applies with $Q_n = \hat{P}_n^Y$. The complete argument is given in Online Appendix 1.

2.7 Scope and Limitations

We explicitly constrain the theoretical claims of this framework. Theorem 10 relies on differentiability of the fixed population projection and on the stated moment conditions. Non-degeneracy is used only for the consequence that transfers projection equivalence to the usual first-order asymptotic distribution of a U -statistic. For highly non-smooth statistics, including indicator-based and rank-based functionals, or for degenerate U -statistics whose limits are governed by higher-order terms, the present first-order analysis need not apply. Furthermore, for degenerate U -statistics (where $g_P(\mathbf{x}_i) = 0$), the first-order Hájek projection vanishes, and the limiting distribution is governed by second-order Gaussian chaos. In such regimes, the surrogate framework makes no first-order preservation claim. Classical nonparametric bootstrap procedures can fail or require statistic-specific modification for important non-smooth and degenerate functionals; the present analysis does not address those cases.

The projection-equivalence result applies to smooth fixed-degree U -statistics under the stated moment and differentiability conditions, without requiring non-degeneracy. The usual first-order distributional consequence for the full U -statistic additionally requires

$$\text{Var}_P\{g_P(\mathbf{X})\} > 0$$

and the stated asymptotic-linear representations. The present analysis does not address higher-order limiting behavior for degenerate U -statistics. Extensions to PCA eigenvalues, learner-specific prediction errors, and other algorithmic functionals require verification of their own differentiability, asymptotic-linearity, and any additional problem-specific conditions.

Whether surrogate preprocessing is useful for a particular downstream objective remains a separate finite-sample and task-specific question.

The theoretical results therefore concern smooth, non-degenerate fixed-degree U-statistics and related functionals for which a suitable asymptotically linear expansion has been independently established. Covariance-based quantities provide natural examples. Extensions to PCA eigenvalues, learner-specific prediction errors, and other algorithmic functionals require verification of their own differentiability, asymptotic-linearity, and problem-specific conditions. Whether surrogate preprocessing is useful for a particular downstream objective remains a separate finite-sample and task-specific question.

3 Algorithmic Consequences and Domain Restrictions

The minimal-distortion result in Section 2 is a fixed-correspondence Euclidean optimization problem, not an optimal-transport problem over arbitrary couplings or permutations of empirical support points. The correspondence

$$x_i \mapsto y_i, \quad i = 1, \dots, n,$$

is prescribed throughout, and the objective is total squared displacement under that correspondence. Accordingly, all optimality claims in this paper refer specifically to this fixed-correspondence point-cloud problem. When deploying the surrogate transformation as a preprocessing layer in general machine-learning pipelines, it is important to distinguish this geometric result from the separate question of whether the transformed representation remains compatible with the data domain and downstream algorithm. This section therefore describes the geometric invariances of the transformation and identifies domain-specific and algorithmic settings in which global covariance-scale calibration may have limited, neutral, or undesirable downstream effects.

Corollary 12 (Nearest-neighbor preservation under consistent feature transformation)

Let the training observations be transformed according to

$$\mathbf{y}_i = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}), \quad s > 0,$$

and transform a query or test covariate $\mathbf{x}_0$ using the same training-sample parameters:

$$\mathbf{y}_0 = \bar{\mathbf{x}} + s(\mathbf{x}_0 - \bar{\mathbf{x}}).$$

Then, for every training observation i ,

$$\|\mathbf{y}_0 - \mathbf{y}_i\|_2 = s\|\mathbf{x}_0 - \mathbf{x}_i\|_2.$$

Consequently, the ordering of distances from the query point to the training observations is preserved. In the absence of distance ties, the identities of the K nearest neighbors are therefore unchanged.

Proof. The stated transformation gives

$$\mathbf{y}_0 - \mathbf{y}_i = s(\mathbf{x}_0 - \mathbf{x}_i),$$

and the conclusion follows because $s > 0$. This result concerns the standard nearest-neighbor ordering underlying K -nearest-neighbor procedures; see Cover and Hart (1967).

3.1 Support Constraints and Bounded Domains

Because the surrogate transformation

$$\mathbf{y}_i = s(\mathbf{x}_i - \bar{\mathbf{x}}) + \bar{\mathbf{x}}, \quad s = \sqrt{\frac{n}{n-1}} > 1,$$

multiplies each observation's distance from the empirical center of mass by s , it strictly increases every nonzero distance from that center. Consequently, the transformation does not generally preserve strict support constraints or integer-valued sample spaces. Accordingly, the transformation should be applied only after confirming that the affine-expanded feature representation remains admissible for the data domain, preprocessing pipeline, and downstream learner.

For bounded data, such as probabilities $x_{ij} \in [0, 1]$, strictly positive variables $x_{ij} > 0$, or categorical and binary indicators, the transformed values y_{ij} may fall outside the admissible support. For example, in a nonconstant Bernoulli sample with $0 < \bar{x}_j < 1$, the transformation maps an observed zero to

$$y_{ij} = (1 - s)\bar{x}_j < 0,$$

and an observed one to

$$y_{ij} = s + (1 - s)\bar{x}_j > 1.$$

Likewise, integer-valued count observations are generally mapped to noninteger real values. These support violations limit the direct use of the affine transformation for strictly bounded, discrete, or constrained data.

3.2 Algorithmic Invariance and Ensemble Selection

Operating recommendation. The relevance of surrogate preprocessing depends on whether the downstream procedure is sensitive to the global affine rescaling induced by the transformation. For scale-invariant procedures, the transformation may leave the principal output unchanged. For scale-sensitive pipelines that operate directly on observations, however, support-level calibration can alter subsequent fitting or resampling operations in ways that are not generally recoverable from a simple post-hoc covariance correction.

Accordingly, surrogate preprocessing is most naturally evaluated as a covariance-scale calibration step for continuous, variance-sensitive procedures. Its use should be assessed relative to the particular learning objective, tuning rule, loss function, and inferential target. Exact calibration of the empirical covariance scale does not imply improved predictive uncertainty, clustering accuracy, covariance-estimation loss, or interval coverage in every pipeline.

The following examples use standard formulations of tree-based methods, support vector machines, nearest-neighbor methods, and K -means clustering; see Hastie et al. (2009) for general background. For an idealized axis-aligned CART splitting rule on continuous numerical predictors, candidate partitions determined solely by within-feature order are unchanged under the surrogate map, because each feature undergoes a strictly increasing affine transformation. Consequently, under matched feature subsampling, randomization, sample weights, stopping rules, threshold conventions, missing-data treatment, and deterministic tie-breaking, the mathematical splitting rule induces the same partition of the training

observations. Implementations that apply feature preprocessing, randomized cutpoints, categorical encodings, numerical tolerances, or implementation-specific missing-data rules require separate verification.

Support vector machines provide a contrasting scale-sensitive example. For a hard-margin linear SVM, the common rescaling can be absorbed by a corresponding rescaling of the separating-hyperplane coefficients, leaving the classification boundary unchanged when test features are transformed consistently. In soft-margin or kernel SVMs, fixed regularization and kernel parameters need not be invariant under surrogate calibration. For the Gaussian RBF kernel

$$K_\gamma(\mathbf{x}, \mathbf{x}') = \exp\{-\gamma\|\mathbf{x} - \mathbf{x}'\|_2^2\},$$

pairwise kernel values are preserved only when

$$\gamma_Y = \frac{\gamma_X}{s^2}.$$

Thus, holding γ numerically fixed deliberately changes the effective kernel scale. More generally, scale-dependent regularized and kernel procedures require corresponding parameter transformation or retuning.

For matched-initialization K -means (Lloyd, 1982), using corresponding bootstrap re-samples and identical deterministic tie-breaking, common positive scaling and translation preserve the sequence of nearest-centroid comparisons and therefore preserve cluster assignments. The centroid coordinates and their bootstrap variability are nevertheless rescaled. For K -nearest-neighbor procedures, nearest-neighbor identities are preserved only when the same positive affine map, defined using the training-sample mean and scale factor, is applied consistently to both the training covariates and all test or query covariates; see Corollary 12. Other components of a complete prediction or resampling pipeline can nevertheless affect downstream uncertainty summaries.

Tie-breaking and numerical implementation. The assignment-invariance statement for matched-initialization K -means is an exact-arithmetic result conditional on the same deterministic tie-breaking rule in the two runs. In finite-precision implementations, however, equality or near-equality of distances can be resolved differently because of rounding, tolerance thresholds, stopping criteria, or implementation-specific update order. Thus, exact agreement in software additionally requires matched numerical tolerances, initialization, resampling indices, stopping rules, and deterministic tie-handling conventions.

What surrogate calibration does not guarantee. Exact covariance calibration does not imply improved predictive accuracy, improved full-covariance estimation, nominal bootstrap coverage for arbitrary statistics, or improved clustering accuracy. The transformation changes global second-moment scale while preserving relative centered geometry. Its downstream effect therefore depends on whether the procedure, its tuning parameters, and the inferential target are sensitive to that scale. The transformation can be beneficial, neutral, or unfavorable when finite-sample error is dominated by mechanisms other than empirical covariance compression.

4 Algorithmic Implementation

The transformation can be implemented through vectorized matrix operations, avoiding explicit element-wise loops. Standard numerical environments typically dispatch these operations to optimized linear-algebra libraries. Its implementation requires only centering, scalar rescaling, and translation of the observed data matrix.

In this section, we present the vectorized matrix formulation of the surrogate transformation. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote the empirical design matrix, where each of the n rows corresponds to a p -dimensional observation. Let $\mathbf{1}_n \in \mathbb{R}^n$ denote a column vector of ones.

4.1 The Preprocessing Algorithm

The transformation is implemented by computing feature means, centering the design matrix, multiplying by the scalar s , and translating back to the original mean. The resulting vectorized formulation avoids explicit observation-by-observation loops.

Algorithm 1 Vectorized Surrogate Population Preprocessing

Require: Empirical design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$

Ensure: Calibrated surrogate matrix $\mathbf{Y} \in \mathbb{R}^{n \times p}$

1: **Initialize:** Check dimensions n and p . Assert $n > 1$.

2: **Compute Column Means:**

$$\bar{\mathbf{x}}^\top \leftarrow \frac{1}{n} \mathbf{1}_n^\top \mathbf{X}$$

(where $\bar{\mathbf{x}}^\top \in \mathbb{R}^{1 \times p}$ is the row vector of feature means)

3: **Compute Expansion Factor:**

$$s \leftarrow \sqrt{\frac{n}{n-1}}$$

4: **Center the Matrix:**

$$\mathbf{X}_c \leftarrow \mathbf{X} - \mathbf{1}_n \bar{\mathbf{x}}^\top$$

5: **Scale and Translate:**

$$\mathbf{Y} \leftarrow s \mathbf{X}_c + \mathbf{1}_n \bar{\mathbf{x}}^\top$$

6: **return** $\mathbf{Y}$

Once $\mathbf{Y}$ is generated, it can be supplied to a downstream resampling or learning procedure in place of $\mathbf{X}$, provided that the transformed data remain compatible with the procedure's data-domain requirements.

Response variables. The surrogate transformation is applied to the feature matrix supplied to the learning or resampling procedure. Response variables are not transformed unless a separate scientific or modeling argument calls for response-scale calibration. In particular, the regression experiments below transform the covariates while retaining the observed responses unchanged.

4.2 Computational Complexity

Computing the feature means and applying the common affine map each require one linear pass over the $n \times p$ design matrix. Therefore, surrogate construction requires $O(np)$ time. An in-place implementation uses $O(p)$ auxiliary storage for the feature-mean vector; retaining a separate surrogate matrix requires $O(np)$ additional storage. This one-time preprocessing step does not change the leading-order cost of downstream procedures that repeatedly fit nontrivial models or explicitly form dense covariance matrices.

4.3 Data-Layer Calibration Versus Analytical Post-Hoc Rescaling

Because the surrogate transformation satisfies

$$\mathbf{S}_{Y,\text{emp}} = \frac{n}{n-1} \mathbf{S}_{X,\text{emp}},$$

a natural benchmark is analytical post-hoc rescaling of a covariance-based output computed from the ordinary empirical support. The two approaches are algebraically equivalent only when the downstream output is determined by the relevant covariance-scale quantity and its response to scalar rescaling is known.

For example, if a procedure returns a covariance matrix computed with denominator n , then multiplying that matrix by $n/(n-1)$ gives exactly the covariance computed from the surrogate support. Likewise, under matched resampling, PCA eigenvalues and percentile-bootstrap interval endpoints are multiplied by $n/(n-1)$, while centroid deviations and centroid variances in the matched-initialization K -means setting are multiplied by $\sqrt{n/(n-1)}$ and $n/(n-1)$, respectively. In these cases, surrogate preprocessing supplies a data-layer realization of an available analytical rescaling.

For pipelines acting directly on observations, equivalence to post-hoc rescaling must be established case by case. It can fail because common affine rescaling changes optimization, regularization, hyperparameter selection, stopping behavior, or the distribution of fitted resample-specific objects. Conversely, it may be recovered under an appropriate transformation of scale-dependent hyperparameters. The practical value of surrogate preprocessing is therefore not universal superiority to post-hoc correction, but the ability to impose the specified covariance scale at the empirical-support level before a compatible pipeline is executed.

4.4 Operational Role of Empirical-Support Calibration

The surrogate construction and covariance-level rescaling should be distinguished carefully. If a downstream procedure depends only on a covariance matrix and its response to scalar multiplication is known, then surrogate preprocessing is algebraically equivalent to applying the factor

$$\frac{n}{n-1}$$

directly to the denominator- n covariance estimator. Likewise, for a matched bootstrap procedure whose resample-level statistic is homogeneous under common positive affine rescaling, surrogate preprocessing is equivalent to applying the corresponding deterministic rescaling to each resample-level output.

Table 2: Relationship between empirical-support calibration and covariance-level post-hoc rescaling. For covariance-based homogeneous procedures, surrogate preprocessing and resample-level covariance rescaling are algebraically equivalent. Differences can arise when downstream procedures operate directly on observations and are sensitive to absolute feature scale.

Representation or operation	Empirical support	Covariance scale	Bootstrap support
Standard empirical support	$\{\mathbf{x}_i\}_{i=1}^n$	$\mathbf{S}_{X,\text{emp}}$	$\{\mathbf{x}_i\}_{i=1}^n$
Surrogate empirical support	$\{\mathbf{y}_i\}_{i=1}^n$	$\mathbf{S}_{Y,\text{emp}} = \frac{n}{n-1} \mathbf{S}_{X,\text{emp}}$	$\{\mathbf{y}_i\}_{i=1}^n$
Post-hoc covariance rescaling	$\{\mathbf{x}_i\}_{i=1}^n$	$\frac{n}{n-1} \mathbf{S}_{X,\text{emp}}$	$\{\mathbf{x}_i\}_{i=1}^n$

The empirical-support representation is operationally relevant when a downstream pipeline consumes observations rather than only a covariance matrix. The standard empirical bootstrap draws from

$$\hat{P}_n^X = \frac{1}{n} \sum_{i=1}^n \delta_{x_i},$$

whereas the surrogate procedure draws from

$$\hat{P}_n^Y = \frac{1}{n} \sum_{i=1}^n \delta_{y_i}.$$

Thus, covariance-scale calibration is imposed before resampling, fitting, or other observation-level operations are performed. Whether this changes a downstream result depends on the algorithm, its hyperparameters, its treatment of absolute feature scale, and the transformation applied to any associated training, test, or query inputs.

Accordingly, the surrogate construction is not proposed as a universally different or superior alternative to covariance-level correction. Rather, it provides a deterministic empirical-support realization of the calibrated global covariance scale. For covariance-derived or positively homogeneous procedures, the two implementations can be algebraically equivalent. For affine-invariant procedures, calibration may have no effect. For scale-sensitive pipelines that operate directly on observations, however, support-level calibration can propagate through subsequent fitting or resampling operations in ways that are not generally represented by a simple post-hoc covariance correction.

5 Empirical Scope Assessment

The empirical study is organized around four distinct questions rather than a single performance criterion. First, we verify exact algebraic identities at the empirical-support level. Second, we examine how the deterministic covariance-scale factor propagates to covariance and spectral quantities. Third, we evaluate whether that propagation changes task-specific statistical losses or uncertainty summaries. Finally, we include settings in which affine invariance implies that no substantive downstream improvement should occur. This organization

is intended to distinguish exact consequences of the proposed transformation from empirical claims about statistical performance.

We evaluate the surrogate population framework across representative computational-statistics and learning tasks involving covariance scale, spectral variation, distance geometry, and regularized fitting. These settings are selected not as a benchmark contest, but to assess when the finite-sample covariance-scale convention of the empirical support is consequential, analytically redundant, or irrelevant to a downstream procedure.

Reproducibility setup. Across all experiments, we compare pipelines using the standard empirical data $\mathbf{X}$ with otherwise identical pipelines using the calibrated surrogate data $\mathbf{Y}$. Experiments 1, 2, and 4, together with the sample-size and dimension-to-sample-size sensitivity analyses, use $M = 1000$ paired Monte Carlo replications. Experiment 3 uses $M = 500$ paired replications. Unless otherwise stated, resampling-based procedures use $B = 500$ bootstrap resamples; the bootstrap PCA analyses use $B = 2000$ resamples per Monte Carlo replication. The numerical verification in Table 3 uses a separate set of $M = 100$ independent Monte Carlo replications, distinct from the replication counts used for the primary experiments. Each named experiment and sensitivity analysis uses its own fixed replication-level random-number stream. Consequently, overlapping parameter configurations appearing in distinct analyses are reproducible within their respective scripts but need not yield identical Monte Carlo summaries across separately seeded analyses.

Data and code availability. All simulation code, experiment configurations, and instructions for reproducing the reported tables and figures are available at <https://github.com/asrivas18/covariance-scale-calibration>. The repository contains the scripts used for the post-hoc verification, Experiments 1–4, the sample-size and dimension-to-sample-size sensitivity analyses, ridge-shrinkage robustness, the gene-expression illustration, and the split-conformal reference analysis. The repository will be frozen in a versioned archival release upon submission.

The release records software dependencies, random-number generation, simulation seeds, computational environment, data-generating mechanisms, and hyperparameter settings used in the experiments. The final accepted release will additionally be archived in Zenodo and assigned a version-specific DOI.

All comparisons between ordinary and surrogate pipelines are paired: within each Monte Carlo replication, the same simulated dataset, bootstrap-resampling index vectors, and other stochastic inputs are used for both pipelines whenever the corresponding procedure permits such matching. Consequently, reported $\mathbf{Y} - \mathbf{X}$ differences isolate the effect of surrogate preprocessing and generally have lower Monte Carlo error than comparisons based on independent simulation runs.

Online Appendix 2 provides a real-data illustration using a moderate-dimensional continuous gene-expression data set. The illustration verifies the exact empirical covariance-scale identity and its deterministic consequences for PCA summaries, but does not claim population-bias correction or inferential improvement from a single observed data set. The empirical study is not intended as a benchmark competition among predictive algorithms or uncertainty-quantification procedures. Its purpose is to verify exact finite-sample propagation identities under matched stochastic inputs and to identify controlled settings in which

covariance-scale calibration is beneficial, neutral, or unfavorable for a specified downstream quantity.

5.1 Analytical Post-Hoc Rescaling Benchmarks

Before reporting task-specific performance, we distinguish outputs for which surrogate preprocessing is exactly reproducible by analytical post-hoc rescaling from outputs for which it is not. For each deterministic identity, the ordinary and surrogate pipelines use matched stochastic inputs. Table 3 reports the maximum absolute discrepancy between the surrogate output and its analytically rescaled ordinary-data counterpart over a separate set of $M = 100$ independent Monte Carlo replications. Values at numerical precision verify the stated equivalences; non-equivalence is assessed directly in the task-specific experiments below.

Table 3: Numerical verification of exact analytical post-hoc rescaling relations over $M = 100$ independent Monte Carlo replications. Entries report maximum absolute discrepancies. The checks use matched stochastic inputs and the matching conditions stated in the corresponding experiments.

Quantity	Analytical relation	Maximum discrepancy
Denominator- n covari- ance	$\mathbf{S}_{Y,\text{emp}} = \frac{n}{n-1} \mathbf{S}_{X,\text{emp}}$	1.33×10^{-15}
Matched PCA eigenval- ues	$\lambda_j(\mathbf{S}_Y) = \frac{n}{n-1} \lambda_j(\mathbf{S}_X)$	2.66×10^{-15}
Matched PCA bootstrap draws	$\lambda_{j,Y}^{*(b)} = \frac{n}{n-1} \lambda_{j,X}^{*(b)}$	6.22×10^{-15}
Matched PCA percentile endpoints	$q_{\alpha,Y} = \frac{n}{n-1} q_{\alpha,X}$	3.55×10^{-15}
Matched K -NN neigh- bor ordering	identical distance ordering un- der matched scaling	0
Matched K -means as- signments	identical assignments under stated matching	0
Matched K -means cen- troid deviations	$\hat{\boldsymbol{\mu}}_Y - \bar{\mathbf{Y}} = \sqrt{\frac{n}{n-1}} (\hat{\boldsymbol{\mu}}_X - \bar{\mathbf{X}})$	8.88×10^{-16}
Matched K -means cen- troid variance	$V_Y = \frac{n}{n-1} V_X$	2.78×10^{-17}

All nonzero discrepancies in Table 3 are at the scale of double-precision floating-point arithmetic. Thus, under the stated matching conditions, the numerical results reproduce the corresponding exact algebraic identities. Table 3 verifies implementation-level identities and invariances under the stated matching conditions. It is not evidence of improved statistical risk, coverage, or prediction performance, which are evaluated separately below.

All empirical comparisons should therefore be interpreted conditionally on the specified data-generating mechanism, sample size, learner, tuning convention, loss, and matching protocol. They do not establish a uniform performance ordering across covariance estimators, prediction procedures, or clustering algorithms.

5.2 Experiment 1: High-Dimensional Covariance Estimation ($p > n$)

Task. We consider a Gaussian design in which observations follow

$$\mathbf{X}_i \sim N_p(0, \Sigma), \quad \Sigma_{jk} = \rho^{|j-k|},$$

with an AR(1)-type covariance structure. We report results for the representative high-dimensional configuration $(n, p) = (30, 100)$ with $\rho = 0.7$, so that $p > n$.

Purpose of the experiment. This experiment is designed as a boundary-of-validity assessment rather than as a demonstration that surrogate preprocessing improves covariance estimation. The exact finite-sample construction replaces the denominator- n empirical-support covariance scale by the denominator- $(n - 1)$ sample-covariance scale of the original observations, but it does not imply optimality under Frobenius loss, improved shrinkage estimation, or improved precision-matrix estimation. We therefore examine whether this exact scale replacement translates into improved matrix-estimation performance under two standard linear-shrinkage procedures. Because the surrogate map globally rescales the centered sample, a scale-equivariant covariance procedure can inherit a deterministic multiplicative relationship between its ordinary and surrogate outputs. The purpose of this experiment is therefore not to propose a new covariance estimator, but to quantify how the prescribed empirical-support scale change interacts with losses evaluated against a fixed population covariance or precision matrix.

We evaluate two covariance-shrinkage procedures: Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004) and Oracle Approximating Shrinkage (OAS) (Chen et al., 2010). In both cases the covariance estimator uses the denominator- n empirical covariance as its input and shrinks toward the scaled-identity target whose trace matches that input. For the reported precision-error metric we compute

$$\hat{\Omega} = (\hat{\Sigma} + 10^{-4} \mathbf{I}_p)^{-1}.$$

The same numerical inversion ridge is used for the ordinary and surrogate estimators within each paired replication.

The surrogate transformation is a deterministic rescaling of the centered sample:

$$\mathbf{Y} = \mathbf{1}_n \bar{\mathbf{x}}^\top + \sqrt{\frac{n}{n-1}} (\mathbf{X} - \mathbf{1}_n \bar{\mathbf{x}}^\top).$$

Consequently, the covariance computed from $\mathbf{Y}$ using denominator n is equal to the usual centered sample covariance computed from $\mathbf{X}$ using denominator $n - 1$. This experiment examines whether that trace-scale correction also affects global covariance and precision-matrix estimation error.

Metrics. For each replication, we compute:

- covariance Frobenius error,

$$\|\hat{\Sigma} - \Sigma\|_F;$$

- precision-matrix Frobenius error,

$$\|\hat{\Omega} - \Sigma^{-1}\|_F,$$

where $\hat{\Omega}$ is obtained by inverting the regularized covariance estimate;

- trace bias,

$$\text{tr}(\hat{\Sigma}) - \text{tr}(\Sigma).$$

We report Monte Carlo means and Monte Carlo standard errors (MCSEs). Because the ordinary and surrogate procedures are evaluated on the same simulated data set in each replication, we also report paired differences, defined as the surrogate metric minus the ordinary-data metric.

Results. Table 4 reports results over $M = 1000$ paired Monte Carlo replications. Under both Ledoit–Wolf and OAS shrinkage, the ordinary-data estimator has a negative trace bias of approximately -3.38 . Applying the same shrinkage operator to the surrogate population moves the mean trace bias close to zero, to approximately -0.05 . The paired trace increase is 3.332 (MCSE 0.005), consistent with the theoretical covariance-scale factor $n/(n-1) = 30/29$.

The surrogate transformation has little effect on covariance Frobenius error: the error increases by 0.021 for Ledoit–Wolf and 0.007 for OAS. Precision Frobenius error increases systematically, by 0.336 for Ledoit–Wolf and 0.330 for OAS. Thus, in this AR(1) design, surrogate rescaling changes the trace scale of the empirical covariance but does not improve global covariance or precision-matrix estimation error.

The agreement between Ledoit–Wolf and OAS indicates that the observed trace-scale change is not specific to a single linear-shrinkage rule. At the same time, the results clarify the scope of the method: changing the finite-sample covariance-normalization convention of the empirical support does not by itself solve the harder high-dimensional problem of estimating the full covariance or precision matrix. This distinction is important for interpreting the proposed method. The surrogate transformation is designed to change the finite-sample covariance-normalization convention of the empirical support, not to minimize an arbitrary matrix loss. Consequently, exact trace-scale calibration and improved covariance estimation are distinct objectives. This boundary-of-validity experiment demonstrates that exact empirical covariance-scale calibration can be achieved without improving general covariance or precision-matrix estimation under the losses and shrinkage rules considered here.

Table 4: Experiment 1 results for $(n, p) = (30, 100)$ and $\rho = 0.7$, based on $M = 1000$ paired Monte Carlo replications. Entries for standard and surrogate data are Monte Carlo means (MCSEs). The final column is the paired difference $Y - X$ (MCSE); negative values favor the surrogate for Frobenius-error metrics. For the precision-error metric, $\hat{\Omega} = (\hat{\Sigma} + 10^{-4}\mathbf{I}_p)^{-1}$ was used for both members of each paired replication.

Estimator	Metric	Standard (X)	Surrogate (Y)	Difference (Y - X)
Ledoit–Wolf	Frob. covariance error	11.18 (0.01)	11.20 (0.01)	0.021 (0.001)
Ledoit–Wolf	Frob. precision error	23.07 (0.01)	23.41 (0.01)	0.336 (< 0.001)
Ledoit–Wolf	Trace bias	-3.38 (0.13)	-0.05 (0.14)	3.332 (0.005)
OAS	Frob. covariance error	11.16 (0.01)	11.16 (0.01)	0.007 (0.001)
OAS	Frob. precision error	23.41 (0.01)	23.74 (0.01)	0.330 (< 0.001)
OAS	Trace bias	-3.38 (0.13)	-0.05 (0.14)	3.332 (0.005)

5.3 Experiment 2: Global Covariance-Scale Propagation in Bootstrap PCA

Task. Principal Component Analysis (PCA) relies on the eigendecomposition of the covariance matrix, and bootstrap procedures are commonly used to quantify uncertainty in eigenvalues representing explained variance. We study bootstrap PCA under a Gaussian design with population eigenvalues

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = (10, 5, 2, 1, 0.5).$$

To avoid a simulation design tied to the coordinate axes, each replication uses a randomly rotated covariance matrix

$$\Sigma = \mathbf{Q} \text{diag}(10, 5, 2, 1, 0.5) \mathbf{Q}^\top,$$

where $\mathbf{Q}$ is an independently generated Haar-distributed orthogonal matrix for each Monte Carlo replication. Thus, the population eigenvalues are fixed while the population eigenvectors vary across replications. In the implementation, $\mathbf{Q}$ is generated from the QR decomposition of a $p \times p$ matrix of independent standard normal variates, with the diagonal sign convention adjusted to obtain a Haar-distributed orthogonal matrix.

For the representative configuration $n = 25$ and $p = 5$, we perform ordinary nonparametric bootstrap PCA on both the original empirical data $\mathbf{X}$ and the surrogate data $\mathbf{Y}$, using $B = 2000$ bootstrap resamples per Monte Carlo replication. PCA covariance matrices are computed using denominator n . We additionally evaluate bias-corrected and accelerated (BCa) intervals (Efron, 1987) computed from the ordinary data $\mathbf{X}$ as a conventional interval benchmark.

Because the surrogate is a global rescaling of the centered sample,

$$\mathbf{Y} - \mathbf{1}_n \bar{\mathbf{x}}^\top = \sqrt{\frac{n}{n-1}} (\mathbf{X} - \mathbf{1}_n \bar{\mathbf{x}}^\top).$$

By Corollary 7, the denominator- n empirical covariance matrices satisfy

$$\hat{\Sigma}_Y = \frac{n}{n-1} \hat{\Sigma}_X.$$

Thus, they have the same eigenspaces, and every corresponding empirical covariance eigenvalue is multiplied by $n/(n-1)$. Under matched bootstrap resampling index vectors, the same relation holds for every paired bootstrap resample. Accordingly, every sample and bootstrap eigenvalue obtained from the surrogate population is exactly the corresponding ordinary eigenvalue multiplied by the deterministic factor $n/(n-1)$. Thus, Experiment 2 is an empirical verification and finite-sample consequence assessment of Corollary 7, rather than an independent eigenvalue-specific correction procedure.

For PCA quantities computed solely from covariance matrices, surrogate preprocessing is therefore exactly equivalent to multiplying the corresponding ordinary-bootstrap covariance matrix by $n/(n-1)$ at each matched bootstrap resample. The PCA experiment should consequently be interpreted as a controlled study of how the calibrated global covariance scale propagates through a resampling-based spectral functional, rather than as evidence that the surrogate representation provides a distinct PCA correction beyond this deterministic rescaling.

More precisely, let $X_c^{*(b)}$ and $Y_c^{*(b)}$ denote the centered data matrices for the b th matched bootstrap resamples from the ordinary and surrogate empirical supports, respectively. If the same bootstrap resampling index vector is used for both supports, then

$$Y_c^{*(b)} = sX_c^{*(b)}, \quad s^2 = \frac{n}{n-1}.$$

Indeed, the bootstrap-resample mean transforms according to

$$\bar{y}^{*(b)} = \bar{x} + s \left(\bar{x}^{*(b)} - \bar{x} \right),$$

so centering each matched resample at its own empirical mean yields $Y_c^{*(b)} = sX_c^{*(b)}$. Therefore, the denominator- n bootstrap covariance matrices satisfy

$$S_{Y,\text{emp}}^{*(b)} = s^2 S_{X,\text{emp}}^{*(b)} = \frac{n}{n-1} S_{X,\text{emp}}^{*(b)}.$$

Consequently, for every bootstrap replicate b and eigenvalue index j ,

$$\lambda_j \left(S_{Y,\text{emp}}^{*(b)} \right) = \frac{n}{n-1} \lambda_j \left(S_{X,\text{emp}}^{*(b)} \right).$$

Interpretation. The PCA analysis is a controlled finite-sample study of exact global scale propagation through a spectral resampling functional. Any coverage change reflects the interaction of the deterministic factor $n/(n-1)$ with the finite-sample location, spread, and bias of an ordered eigenvalue relative to its population target. The analysis does not claim an intrinsic computational advantage of surrogate preprocessing over the corresponding matched post-hoc rescaling.

Metrics. We focus on the leading two population eigenvalues, $(\lambda_1, \lambda_2) = (10, 5)$. For each Monte Carlo replication, we compute:

- the ordinary sample eigenvalues from $\mathbf{X}$ and $\mathbf{Y}$;
- the means of the bootstrap eigenvalue distributions;
- empirical coverage and width of 95% percentile bootstrap intervals based on $\mathbf{X}$ and $\mathbf{Y}$;
- empirical coverage and width of 95% BCa intervals based on $\mathbf{X}$.

All reported entries are Monte Carlo means with Monte Carlo standard errors (MCSEs), based on $M = 1000$ paired Monte Carlo replications. Coverage is assessed with respect to the fixed population eigenvalues. Interval-width results are reported in Online Appendix 2.

Results. For the leading eigenvalue, the ordinary sample mean is 10.21, already above the population value 10, while the surrogate sample mean is 10.64. Thus, applying the global scale transformation moves the already upward-biased leading eigenvalue farther above its population value. The same pattern appears in the bootstrap means, which increase from 10.37 for $\mathbf{X}$ to 10.80 for $\mathbf{Y}$; see Table 5.

For the second eigenvalue, the ordinary sample mean is 4.67, below the population value 5, whereas the surrogate sample mean is 4.87. Hence, the same global rescaling partially

offsets the downward bias of λ_2 . The bootstrap mean remains lower than the population value for both procedures, at 4.26 for $\mathbf{X}$ and 4.44 for $\mathbf{Y}$.

Table 6 reports interval coverage. For λ_1 , ordinary and surrogate percentile intervals have observed coverage of 93.8% and 95.1%, respectively. The paired coverage difference of 1.3 percentage points (MCSE 0.5 percentage points) is a finite-sample consequence of the deterministic scale change. The BCa benchmark has lower coverage, 88.4%, in this configuration.

For λ_2 , ordinary percentile intervals under-cover the population value, achieving 83.2% coverage. Surrogate rescaling increases observed coverage to 87.4%, a paired gain of 4.2 percentage points (MCSE 0.7 percentage points), but the result remains below nominal coverage. Because the surrogate transformation deterministically rescales the entire bootstrap eigenvalue distribution, the observed coverage increase should be interpreted as a scale-induced finite-sample effect rather than evidence that surrogate preprocessing generally corrects eigenvalue-specific bootstrap bias. BCa intervals achieve 86.5% coverage and do not provide a uniform improvement in this setting.

The increase in observed coverage for λ_2 is accompanied by a predictable increase in interval width. Because the surrogate multiplies every bootstrap eigenvalue by $n/(n-1)$, it also multiplies each percentile interval width by the same factor. For $n = 25$, surrogate percentile intervals are therefore approximately 4.17% wider than ordinary percentile intervals; see Online Appendix 2. The mean width increases from 9.68 to 10.08 for λ_1 and from 4.09 to 4.26 for λ_2 . This coverage-width change should be interpreted as a scale-induced finite-sample effect rather than as a general bootstrap-calibration guarantee.

Overall, surrogate preprocessing exactly implements the specified global empirical covariance-scale transformation and, in this design, increases the observed bootstrap coverage for the downward-biased second eigenvalue. However, it does not correct eigenvalue-specific finite-sample bias, does not materially affect leading-eigenvalue coverage, and does not provide a general guarantee of nominal bootstrap PCA interval coverage.

The PCA experiment therefore demonstrates the propagation of the deterministic global covariance-scale transformation through a nonlinear spectral functional. The surrogate and covariance-rescaled implementations are equivalent at the matched-resample level, while the resulting coverage changes illustrate that a deterministic covariance-scale transformation can nevertheless alter a nonlinear inferential procedure in finite samples. The experiment is thus evidence for the propagation of covariance-scale calibration, rather than evidence of an intrinsic computational advantage of surrogate preprocessing for PCA.

Table 5: Experiment 2 eigenvalue summaries for $n = 25$, $p = 5$, $B = 2000$, and $M = 1000$. Entries are Monte Carlo means (MCSEs).

Eigenvalue	Population	Sample: X	Sample: Y	Bootstrap Mean: X	Bootstrap Mean: Y
λ_1	10.0	10.21 (0.08)	10.64 (0.09)	10.37 (0.08)	10.80 (0.08)
λ_2	5.0	4.67 (0.04)	4.87 (0.04)	4.26 (0.03)	4.44 (0.03)

Table 6: Experiment 2 empirical 95% interval coverage for $n = 25$, $p = 5$, $B = 2000$, and $M = 1000$. Entries are coverage (MCSE). The final two columns are paired coverage differences, in percentage points, relative to the ordinary percentile interval based on X .

Eigenvalue	Percentile: X	Percentile: Y	BCa: X	$Y - X$	BCa-Percentile: X
λ_1	93.8% (0.8%)	95.1% (0.7%)	88.4% (1.0%)	1.3% (0.5%)	-5.4% (0.7%)
λ_2	83.2% (1.2%)	87.4% (1.0%)	86.5% (1.1%)	4.2% (0.7%)	3.3% (1.2%)

5.4 Sample-Size Sensitivity

The magnitude of the deterministic covariance-scale correction is

$$s^2 - 1 = \frac{1}{n-1}, \quad s^2 = \frac{n}{n-1}.$$

Its direct effect on covariance-scaled quantities therefore decreases as the sample size increases. To assess this behavior, we repeat the covariance and bootstrap PCA experiments over

$$n \in \{10, 20, 30, 50, 100, 250\}.$$

For the covariance analysis, we hold $p = 100$ and $\rho = 0.7$ fixed. Thus, the grid spans the high-dimensional regime ($p > n$) for $n < 100$, the square regime ($p = n$) at $n = 100$, and the classical regime ($p < n$) at $n = 250$. This design is intended to isolate sample-size sensitivity rather than to hold the ratio p/n fixed. We report the trace bias of the denominator- n empirical covariance estimator and the trace ratio

$$\frac{\text{tr}(\mathbf{S}_Y)}{\text{tr}(\mathbf{S}_X)}.$$

By construction, this trace ratio is theoretically equal to $n/(n-1)$, up to floating-point precision. For any covariance estimator $\mathbf{S}$, trace bias is defined relative to the population covariance as

$$\text{tr}(\mathbf{S}) - \text{tr}(\mathbf{\Sigma}).$$

For bootstrap PCA, we retain the population eigenspectrum

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = (10, 5, 2, 1, 0.5)$$

and evaluate the surrogate-to-standard ratios of the first two sample eigenvalues, together with percentile-bootstrap coverage for the second population eigenvalue. Under the surrogate transformation, the eigenvalue ratios are theoretically equal to $n/(n-1)$. In contrast, percentile-bootstrap coverage is a nonlinear inferential quantity and is not expected to obey a deterministic scaling law. Coverage and trace-bias entries are Monte Carlo means with MCSEs based on $M = 1000$ replications. The covariance-scale and surrogate-to-standard eigenvalue ratios are deterministic conditional on each simulated sample and equal $n/(n-1)$ up to floating-point precision. These deterministic ratios are the direct finite-sample implication of Corollary 7; the coverage results, by contrast, are nonlinear inferential quantities and need not obey a deterministic scaling law. The PCA results use $B = 2000$ bootstrap resamples per replication.

Table 7: Sample-size verification of the exact empirical covariance-scale identity in the AR(1) design with $p = 100$ and $\rho = 0.7$. Trace-bias entries are Monte Carlo means (MCSEs) over $M = 1000$ replications. The trace ratio is deterministic conditional on each simulated sample and equals $n/(n - 1)$ up to floating-point precision; consequently, no MCSE is reported for that column.

n	$n/(n - 1)$	Trace Bias: $\mathbf{X}$	Trace Bias: $\mathbf{Y}$	$\text{tr}(\mathbf{S}_Y)/\text{tr}(\mathbf{S}_X)$
10	1.11111	-10.007 (0.226)	-0.008 (0.251)	1.11111
20	1.05263	-4.811 (0.162)	0.199 (0.170)	1.05263
30	1.03448	-3.525 (0.141)	-0.198 (0.145)	1.03448
50	1.02041	-1.838 (0.106)	0.165 (0.108)	1.02041
100	1.01010	-0.857 (0.078)	0.145 (0.078)	1.01010
250	1.00402	-0.353 (0.048)	0.047 (0.048)	1.00402

Results. Table 7 verifies the exact covariance-scale identity across sample sizes. For every Monte Carlo replication, the trace ratio

$$\frac{\text{tr}(\mathbf{S}_Y)}{\text{tr}(\mathbf{S}_X)}$$

equals $n/(n - 1)$ up to floating-point precision. Accordingly, the negative trace bias of the denominator- n covariance estimator decreases from approximately -10 at $n = 10$ to approximately -0.35 at $n = 250$, while the surrogate trace bias remains close to zero.

Table 8: Sample-size sensitivity of bootstrap PCA for the population eigenspectrum $(10, 5, 2, 1, 0.5)$. The surrogate-to-standard eigenvalue ratios are deterministic conditional on each simulated sample and equal $n/(n - 1)$ up to numerical precision; no MCSEs are reported for those columns. Coverage is shown for the second population eigenvalue, $\lambda_2 = 5$, and the final column reports the paired coverage difference.

n	$n/(n - 1)$	$\lambda_1: Y/X$	$\lambda_2: Y/X$	Coverage: X	Coverage: Y	$Y - X$
10	1.11111	1.11111	1.11111	57.3% (1.6%)	68.5% (1.5%)	11.2% (1.0%)
20	1.05263	1.05263	1.05263	79.7% (1.3%)	84.3% (1.2%)	4.6% (0.7%)
30	1.03448	1.03448	1.03448	82.3% (1.2%)	86.1% (1.1%)	3.8% (0.6%)
50	1.02041	1.02041	1.02041	89.1% (1.0%)	90.9% (0.9%)	1.8% (0.4%)
100	1.01010	1.01010	1.01010	92.3% (0.8%)	92.5% (0.8%)	0.2% (0.1%)
250	1.00402	1.00402	1.00402	94.6% (0.7%)	95.0% (0.7%)	0.4% (0.2%)

Table 8 shows that the practical effect on bootstrap PCA coverage is sample-size dependent. The surrogate-to-standard eigenvalue ratio equals $n/(n - 1)$ up to numerical precision, whereas percentile-bootstrap coverage does not obey a deterministic scaling law. For the downward-biased second eigenvalue, surrogate preprocessing increases the observed percentile-interval coverage by 11.2 percentage points at $n = 10$, 4.6 at $n = 20$, 3.8 at $n = 30$, and 1.8 at $n = 50$. These finite-sample changes do not imply nominal coverage or an eigenvalue-specific bias correction. At $n = 100$ and $n = 250$, the observed coverage increases

are only 0.2 and 0.4 percentage points, respectively, with MCSEs of 0.1 and 0.2 percentage points. Thus, the deterministic covariance-scale correction decreases exactly as $1/(n-1)$, whereas the corresponding empirical coverage differences need not decrease monotonically and should be interpreted as finite-sample, task-specific consequences of that scaling. Monte Carlo variation becomes increasingly relevant when the correction itself is small.

5.5 Dimension-to-Sample-Size Sensitivity

A single high-dimensional configuration cannot establish that the practical consequences of the covariance-scale correction are uniform over different dimension-to-sample-size ratios. We therefore examine two complementary designs. Panel A fixes $n = 100$ and varies the dimension according to

$$p \in \{25, 50, 100, 200, 500\}.$$

Panel B fixes $p = 100$ and varies the sample size according to

$$n \in \{25, 50, 100, 200, 500\}.$$

The common configuration $(n, p) = (100, 100)$ links the two panels. The configurations with $p/n > 1$ correspond to the genuinely high-dimensional regime, in which the unregularized empirical covariance matrix is rank deficient.

For each configuration, data are generated from a Gaussian AR(1) model with

$$\Sigma_{jk} = \rho^{|j-k|}, \quad \rho = 0.7.$$

We compare the denominator- n empirical covariance estimator $\mathbf{S}_X$ with its surrogate counterpart

$$\mathbf{S}_Y = \left\{ \frac{n}{n-1} \right\} \mathbf{S}_X.$$

Consequently, for every simulated sample,

$$\frac{\text{tr}(\mathbf{S}_Y)}{\text{tr}(\mathbf{S}_X)} = \frac{n}{n-1},$$

up to floating-point precision. Thus, the multiplicative covariance-scale correction is exactly determined by n and is independent of p .

Although the multiplicative correction is dimension-free, its absolute effect on the covariance trace depends on both dimension and sample size. Since the AR(1) covariance has unit diagonal entries,

$$\text{tr}(\Sigma) = p,$$

and the denominator- n estimator satisfies

$$\mathbb{E}\{\text{tr}(\mathbf{S}_X)\} = \frac{n-1}{n}p,$$

the expected paired trace-bias difference induced by the surrogate is

$$\mathbb{E}[\text{tr}(\mathbf{S}_Y) - \text{tr}(\mathbf{S}_X)] = \frac{p}{n}.$$

Hence, at fixed n , the multiplicative correction remains unchanged across dimensions, whereas its expected absolute trace correction increases linearly with p .

We evaluate trace bias,

$$\text{tr}(\mathbf{S}) - \text{tr}(\mathbf{\Sigma}),$$

relative covariance-estimation error,

$$\text{RE}_{\Sigma}(\hat{\mathbf{\Sigma}}) = \frac{\|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_F}{\|\mathbf{\Sigma}\|_F},$$

and relative precision-estimation error,

$$\text{RE}_{\Omega}(\hat{\mathbf{\Omega}}) = \frac{\|\hat{\mathbf{\Omega}} - \mathbf{\Omega}\|_F}{\|\mathbf{\Omega}\|_F}, \quad \mathbf{\Omega} = \mathbf{\Sigma}^{-1}.$$

Because the empirical covariance matrix is singular when $p \geq n$, the precision estimator is obtained from the ridge-shrinkage covariance estimator

$$\hat{\mathbf{\Sigma}}_{\alpha} = (1 - \alpha)\mathbf{S} + \alpha \left\{ \frac{\text{tr}(\mathbf{S})}{p} \right\} \mathbf{I}_p, \quad \alpha = 0.10,$$

followed by

$$\hat{\mathbf{\Omega}}_{\alpha} = \hat{\mathbf{\Sigma}}_{\alpha}^{-1}.$$

The same fixed shrinkage parameter is used for $\mathbf{S}_X$ and $\mathbf{S}_Y$ within each replication. Because this ridge functional is homogeneous of degree one, the exact relation

$$\mathbf{S}_Y = \frac{n}{n-1} \mathbf{S}_X$$

implies

$$\hat{\mathbf{\Sigma}}_{\alpha,Y} = \frac{n}{n-1} \hat{\mathbf{\Sigma}}_{\alpha,X}, \quad \hat{\mathbf{\Omega}}_{\alpha,Y} = \frac{n-1}{n} \hat{\mathbf{\Omega}}_{\alpha,X}.$$

Thus, under the fixed ridge-shrinkage rule, the surrogate induces a deterministic reciprocal rescaling of the corresponding regularized precision estimate.

All reported trace-bias and error entries are Monte Carlo means with MCSEs in parentheses, based on $M = 1000$ paired replications. The final column reports the paired difference, $Y - X$. The trace ratio is deterministic conditional on each simulated sample and is therefore reported without an MCSE.

Results. Tables 9 and 10 separate the exact covariance-scale identity from dimension-dependent estimation performance. Holding $n = 100$ fixed, the trace ratio equals $100/99 = 1.01010$ for every value of p , confirming that the multiplicative correction is independent of dimension. The mean paired trace-bias difference increases from approximately 0.25 at $p = 25$ to 5.00 at $p = 500$, consistent with the exact expectation p/n up to Monte Carlo error. Thus, the exact scale correction is dimension-free multiplicatively, whereas its absolute trace effect is not.

Relative covariance error increases substantially with p/n . In Panel A, the standard-estimator covariance error rises from 0.2956 at $p/n = 0.25$ to 1.1796 at $p/n = 5$. Surrogate

Table 9: Panel A: dimension-to-sample-size sensitivity with fixed $n = 100$ and varying p . Trace bias is reported on the covariance-trace scale; covariance and precision errors are relative Frobenius-norm errors. For every row, the deterministic trace ratio is $\text{tr}(\mathbf{S}_Y)/\text{tr}(\mathbf{S}_X) = 100/99 = 1.01010$. Precision-error comparisons are specific to the fixed ridge-shrinkage parameter $\alpha = 0.10$ and should not be interpreted as a universal precision-estimation result.

n	p	p/n	Metric	X	Y	$Y - X$
100	25	0.25	Trace bias	-0.235 (0.035)	0.015 (0.036)	0.250 (0.000)
			Covariance error	0.2956 (0.0010)	0.2962 (0.0010)	0.0007 (0.0001)
			Precision error	0.3262 (0.0005)	0.3275 (0.0005)	0.0013 (< 0.0001)
100	50	0.50	Trace bias	-0.529 (0.053)	-0.029 (0.054)	0.500 (0.001)
			Covariance error	0.3954 (0.0007)	0.3976 (0.0007)	0.0022 (< 0.0001)
			Precision error	0.4646 (0.0006)	0.4605 (0.0006)	-0.0041 (< 0.0001)
100	100	1.00	Trace bias	-0.919 (0.075)	0.082 (0.076)	1.001 (0.001)
			Covariance error	0.5427 (0.0006)	0.5469 (0.0006)	0.0042 (< 0.0001)
			Precision error	0.8349 (0.0009)	0.8235 (0.0009)	-0.0114 (< 0.0001)
100	200	2.00	Trace bias	-2.172 (0.108)	-0.173 (0.109)	1.998 (0.001)
			Covariance error	0.7531 (0.0005)	0.7597 (0.0005)	0.0067 (< 0.0001)
			Precision error	1.3879 (0.0011)	1.3687 (0.0011)	-0.0192 (< 0.0001)
100	500	5.00	Trace bias	-5.093 (0.171)	-0.093 (0.172)	4.999 (0.002)
			Covariance error	1.1796 (0.0004)	1.1909 (0.0004)	0.0113 (< 0.0001)
			Precision error	1.8207 (0.0009)	1.7960 (0.0009)	-0.0247 (< 0.0001)

Table 10: Panel B: dimension-to-sample-size sensitivity with fixed $p = 100$ and varying n . Trace bias is reported on the covariance-trace scale; covariance and precision errors are relative Frobenius-norm errors. The trace ratio is deterministic conditional on each simulated sample and equals $n/(n - 1)$. Precision-error comparisons are specific to the fixed ridge-shrinkage parameter $\alpha = 0.10$ and should not be interpreted as a universal precision-estimation result.

n	p	p/n	Metric	X	Y	$Y - X$
25	100	4.00	Trace bias	-4.421 (0.146)	-0.439 (0.152)	3.982 (0.006)
			Covariance error	1.0543 (0.0017)	1.0946 (0.0018)	0.0403 (0.0001)
			Precision error	1.8621 (0.0039)	1.7621 (0.0037)	-0.1001 (0.0002)
50	100	2.00	Trace bias	-1.993 (0.107)	0.007 (0.109)	2.000 (0.002)
			Covariance error	0.7582 (0.0010)	0.7716 (0.0010)	0.0135 (0.0001)
			Precision error	1.4216 (0.0022)	1.3825 (0.0022)	-0.0391 (0.0001)
100	100	1.00	Trace bias	-0.989 (0.078)	0.011 (0.078)	1.000 (0.001)
			Covariance error	0.5422 (0.0005)	0.5464 (0.0005)	0.0042 (< 0.0001)
			Precision error	0.8355 (0.0009)	0.8241 (0.0009)	-0.0114 (< 0.0001)
200	100	0.50	Trace bias	-0.465 (0.053)	0.035 (0.054)	0.500 (< 0.0001)
			Covariance error	0.3882 (0.0004)	0.3893 (0.0004)	0.0011 (0.0000)
			Precision error	0.4547 (0.0003)	0.4528 (0.0003)	-0.0019 (< 0.0001)
500	100	0.20	Trace bias	-0.253 (0.034)	-0.053 (0.034)	0.200 (< 0.0001)
			Covariance error	0.2537 (0.0002)	0.2537 (0.0002)	-0.0000 (< 0.0001)
			Precision error	0.3044 (0.0001)	0.3050 (0.0001)	0.0006 (< 0.0001)

preprocessing produces a small increase in relative Frobenius error throughout this grid, ranging from 0.0007 at $p/n = 0.25$ to 0.0113 at $p/n = 5$. Thus, changing the denominator- n trace-bias convention does not imply a reduction in general covariance-estimation loss. The surrogate provides exact finite-sample scale calibration, but it is not designed to uniformly minimize arbitrary covariance-estimation loss functions.

The regularized precision results depend on the dimension-to-sample-size ratio and on the specified ridge-shrinkage functional. In Panel A, the surrogate has lower relative precision error for $p/n \geq 0.5$, but slightly higher precision error at $p/n = 0.25$. In Panel B, the surrogate has lower relative precision error for $p/n \geq 0.5$, while producing a small increase at $p/n = 0.2$. These precision-error patterns are properties of the specified ridge regularization functional with $\alpha = 0.10$; they do not establish a general ordering of precision estimators. The purpose of this analysis is to demonstrate that exact covariance-scale calibration of the empirical support and statistical risk of a downstream regularized estimator are distinct objects whose relationship depends on p/n , the regularization rule, and the loss function.

Online Appendix 2 examines sensitivity to $\alpha \in \{0.05, 0.10, 0.20\}$ at three representative dimension-to-sample-size configurations. The sign of the paired precision-error difference varies across ridge parameters in some regimes, further confirming that no general precision-risk ordering follows from the exact covariance-scale identity alone.

Overall, the results distinguish two effects. The covariance correction is an exact, dimension-free multiplicative calibration determined by n , whereas its downstream consequences for covariance and precision estimation depend on the loss function, the regularization rule, and the p/n regime.

5.6 Experiment 3: Predictive-Uncertainty Boundary of Validity

Task. We evaluate whether surrogate preprocessing affects predictive uncertainty for two scale-sensitive regression learners: Ridge regression, whose effective regularization depends on feature scale, and K -nearest neighbors (K -NN), whose predictions depend on Euclidean distances. Predictors are generated from a Gaussian design in $p = 10$ dimensions, with responses generated according to

$$y = \beta^\top \mathbf{x} + \varepsilon, \quad \varepsilon \sim N(0, 0.5^2),$$

where β is normalized to have unit Euclidean norm. Each Monte Carlo replication uses $n = 30$ training observations and an independent test set of size $n_{\text{test}} = 100$. For surrogate-based prediction, the surrogate training population is constructed once from the original training sample before bootstrap resampling. Let

$$s_{\text{train}} = \sqrt{\frac{n_{\text{train}}}{n_{\text{train}} - 1}}$$

and let $\bar{\mathbf{x}}_{\text{train}}$ denote the original training-sample mean. The test covariates are transformed using these same training-derived quantities:

$$\mathbf{Y}_{\text{test}} = \mathbf{1}_{n_{\text{test}}} \bar{\mathbf{x}}_{\text{train}}^\top + s_{\text{train}} \left(\mathbf{X}_{\text{test}} - \mathbf{1}_{n_{\text{test}}} \bar{\mathbf{x}}_{\text{train}}^\top \right).$$

The test-set mean is not used to define this transformation. Matched bootstrap training samples are subsequently drawn using the same index vectors from the fixed ordinary and surrogate training populations.

Purpose of the experiment. This experiment delineates a boundary of validity for covariance-scale calibration in predictive uncertainty quantification. The surrogate transformation changes the global covariance scale of the feature representation, but bootstrap quantiles of bagged predictions are not automatically calibrated prediction intervals. We therefore assess whether the deterministic feature-scale adjustment changes prediction error, interval width, or empirical coverage for two representative scale-sensitive regression learners.

For each replication, we construct $B = 500$ bootstrap prediction distributions using Bagged Ridge regression with penalty $\lambda = 1$ and Bagged K -NN with $K = 3$. We compare bootstrap prediction-quantile intervals obtained from the ordinary data $\mathbf{X}$, the surrogate data $\mathbf{Y}$, and post-hoc widened ordinary bootstrap intervals. The post-hoc intervals multiply the ordinary bootstrap half-width by

$$\sqrt{\frac{n}{n-1}} = \sqrt{\frac{30}{29}}.$$

The ordinary bootstrap mean prediction is retained as the center of each post-hoc interval.

Metrics. For each bootstrap procedure, we report:

- test mean squared error (MSE) of the point predictor;
- mean width of the nominal 95% bootstrap prediction-quantile interval;
- empirical test-set coverage of that interval.

The intervals are quantiles of the bootstrap distribution of fitted predictions. They are therefore interpreted as bootstrap prediction-quantile intervals rather than as automatically calibrated prediction intervals. All reported values are Monte Carlo means with Monte Carlo standard errors (MCSEs), based on $M = 500$ paired replications.

Results. Consistent with this boundary-of-validity design, Table 11 shows that surrogate preprocessing does not resolve the undercoverage of bootstrap prediction-quantile intervals. For Bagged Ridge, the ordinary and surrogate procedures have nearly identical test MSEs, 0.38 under both $\mathbf{X}$ and $\mathbf{Y}$, and nearly identical interval widths, 1.54 and 1.55, respectively. Empirical coverage is likewise similar: 75.8% for $\mathbf{X}$ and 76.0% for $\mathbf{Y}$. Post-hoc widening increases mean width to 1.57 and coverage to 76.8%. The observed undercoverage is not in conflict with Theorem 11, which is an asymptotic conditional-bootstrap transfer statement and does not assert finite-sample predictive-interval calibration.

Bagged K -NN exhibits more pronounced undercoverage. The ordinary and surrogate procedures have identical test MSEs of 0.83, identical interval widths of 1.79, and identical empirical coverages of 65.1%. Post-hoc widening increases mean width to 1.82 and coverage to 65.9%. This invariance is expected because a common positive scaling and translation of both training and test covariates preserve all K -NN neighbor orderings.

Overall, surrogate preprocessing and post-hoc interval-width expansion have negligible effects on the undercoverage of bootstrap prediction-quantile intervals in this design. Changing the global covariance scale of the feature representation does not substitute for a procedure specifically designed to target predictive interval coverage. For Ridge regression, surrogate preprocessing changes the covariate scale relative to the fixed penalty parameter.

The near-identical performance in this experiment is therefore an empirical result for the specified sample size, penalty, signal-to-noise ratio, and design distribution; it is not a general invariance property of Ridge regression.

Table 11: Experiment 3 predictive performance for $n = 30$, $p = 10$, $B = 500$, $M = 500$ paired Monte Carlo replications, and $n_{\text{test}} = 100$. Entries are Monte Carlo means with MCSEs in parentheses. Bootstrap intervals are quantiles of the distributions of bootstrap fitted predictions.

Algorithm	Metric	Bootstrap: X	Bootstrap: Y	Post-hoc: X
Bagged Ridge	Test MSE	0.38 (0.00)	0.38 (0.00)	–
	95% interval width	1.54 (0.01)	1.55 (0.01)	1.57 (0.01)
	95% empirical coverage	75.8% (0.4%)	76.0% (0.4%)	76.8% (0.4%)
Bagged K -NN	Test MSE	0.83 (0.01)	0.83 (0.01)	–
	95% interval width	1.79 (0.01)	1.79 (0.01)	1.82 (0.01)
	95% empirical coverage	65.1% (0.4%)	65.1% (0.4%)	65.9% (0.4%)

5.7 Experiment 4: Affine Invariance and Centroid Variability in Bootstrap K -Means

Task. We examine the effect of surrogate rescaling on bootstrap K -means clustering (Lloyd, 1982) in a design with $K = 3$ isotropic Gaussian clusters in $p = 2$ dimensions and 25 observations per cluster. The cluster centers form an equilateral triangle with moderate overlap. This is a structural verification rather than a predictive-performance comparison: under matched initial centroids, corresponding bootstrap resamples, and identical deterministic tie-breaking, common positive scaling and translation preserve the sequence of nearest-centroid comparisons. Thus, no improvement in clustering assignments or ARI is expected; the experiment verifies the predicted rescaling of centroid variability. Without matched transformed initialization, an implementation may select different local optima, so the exact assignment identity need not hold.

Unlike the preceding experiments, the primary purpose here is not to show improved clustering accuracy. Rather, this experiment verifies the exact consequence of global affine rescaling for a distance-based algorithm. For each bootstrap sample, we run K -means on both $\mathbf{X}$ and $\mathbf{Y}$ using matched initial centroids: the initial centers for $\mathbf{Y}$ are the surrogate transformations of the initial centers for $\mathbf{X}$. Corresponding initial centers are supplied in the same ordered cluster positions in the two runs. Under the matched-initialization and deterministic tie-breaking conditions stated below, this order induces a common cluster-label correspondence, so assignment agreement is evaluated without post-hoc label matching. Because centroid deviations are linear in the transformed covariates, their magnitudes are multiplied by

$$s = \sqrt{\frac{n}{n-1}}.$$

Consequently, centroid variances are multiplied by

$$s^2 = \frac{n}{n-1}.$$

We use $B = 500$ bootstrap resamples within each of $M = 1000$ Monte Carlo replications.

Metrics. For each bootstrap resample, we compute:

- the Adjusted Rand Index (ARI) between the bootstrap cluster assignments and the corresponding ground-truth cluster labels;
- the standard deviation of bootstrap ARI values within a Monte Carlo replication;
- agreement between the matched $\mathbf{X}$ and $\mathbf{Y}$ cluster assignments;
- the total centroid variance,

$$\sum_{k=1}^K \text{tr} \left\{ \widehat{\text{Var}}(\hat{\boldsymbol{\mu}}_k) \right\},$$

computed across bootstrap resamples;

- the ratio of surrogate to ordinary total centroid variance.

All entries in Table 12 are Monte Carlo means with Monte Carlo standard errors (MCSEs).

Results. Table 12 confirms the expected matched-initialization affine invariance of K -means assignments. The mean bootstrap ARI is identical for the ordinary and surrogate populations, at 0.829, and the bootstrap ARI standard deviation is likewise identical, at 0.139. Under the common label correspondence induced by the matched ordered initialization, the matched cluster assignments agree exactly across the two representations, with mean agreement equal to 1.0000.

The surrogate changes the numerical coordinates of the centroids but not the cluster assignments. The total centroid variance increases from 16.945 under $\mathbf{X}$ to 17.174 under $\mathbf{Y}$. Their ratio is 1.0135, equal to the theoretical value

$$\frac{n}{n-1} = \frac{75}{74} \approx 1.0135.$$

This equality follows because the surrogate multiplies centered coordinates, and hence centroid deviations across matched bootstrap samples, by

$$s = \sqrt{\frac{n}{n-1}}.$$

Because variance is quadratic in these deviations, total centroid variance is multiplied by

$$s^2 = \frac{n}{n-1}.$$

Thus, surrogate rescaling preserves K -means partitions, ground-truth ARI, and bootstrap variation in clustering accuracy under matched initialization. It only rescales centroid-location variability by the known deterministic factor. This experiment establishes a limitation as well as an identity: global covariance-scale correction does not improve a clustering method whose assignments are invariant to common translation and positive rescaling.

Table 12: Experiment 4 matched-initialization bootstrap K -means verification for $n = 75$, $K = 3$, $p = 2$, $B = 500$, and $M = 1000$. Entries are Monte Carlo means (MCSEs). The near-zero MCSE for the centroid-variance ratio reflects the exact deterministic affine relationship between matched $\mathbf{X}$ and $\mathbf{Y}$ bootstrap clusterings.

Metric	Theoretical Target	Standard (X)	Surrogate (Y)
Mean bootstrap ARI	–	0.829 (0.002)	0.829 (0.002)
Bootstrap ARI SD	–	0.139 (0.000)	0.139 (0.000)
Matched assignment agreement	1.0000	1.0000 (0.0000)	
Total centroid variance	–	16.945 (0.049)	17.174 (0.050)
Centroid-variance ratio Y/X	1.0135	1.0135 (0.0000)	

Taken together, the experiments distinguish exact covariance-scale consequences from task-specific statistical effects. The surrogate transformation replaces the denominator- n trace scale of the empirical support by the denominator- $(n - 1)$ sample-covariance trace scale. This scale change propagates deterministically to covariance eigenvalues and matched K -means centroid variability, while its effects on covariance loss, precision loss, bootstrap coverage, and predictive uncertainty depend on the downstream procedure, regularization rule, inferential target, and loss function. The empirical evidence therefore supports a global covariance-scale calibration interpretation rather than a claim of uniform improvement across machine-learning tasks.

6 Discussion and Extensions

The surrogate construction provides an exact finite-sample representation of the denominator- $(n - 1)$ sample-covariance scale at the level of a transformed empirical support. It is a covariance-scale calibration identity, not a universal statistical correction. For

$$s = \sqrt{\frac{n}{n-1}}, \quad \mathbf{y}_i = \bar{\mathbf{x}} + s(\mathbf{x}_i - \bar{\mathbf{x}}),$$

the transformed empirical support preserves the sample mean and satisfies

$$\mathbf{S}_{Y,\text{emp}} = \frac{n}{n-1} \mathbf{S}_{X,\text{emp}} = \mathbf{S}_X.$$

This identity is algebraic, requires no distributional assumption, and can be implemented in $\mathcal{O}(np)$ time. Under the structural conditions of Section 2, the map is uniquely characterized within the stated coordinate-wise affine class and is the unique minimum-distortion transformation under the prescribed pointwise correspondence and covariance constraint.

Calibration is not estimation. A central distinction throughout this paper is between calibrating the covariance scale of an empirical population and estimating a population covariance matrix. The former is an exact finite-sample algebraic operation:

$$S_{Y,\text{emp}} = \frac{n}{n-1} S_{X,\text{emp}} = S_X.$$

The latter is a statistical estimation problem whose risk depends on sampling variation, dimension, covariance structure, regularization, and the chosen loss function. Consequently, exact covariance-scale calibration can replace the denominator- n empirical-support scale by the denominator- $(n - 1)$ sample-covariance scale while leaving unchanged, or even increasing, the loss of a general covariance or precision estimator. The covariance and precision experiments in Sections 5.2 and 5.5 illustrate this distinction directly.

Finite-sample character of the calibration. As $n \rightarrow \infty$,

$$s - 1 = \sqrt{\frac{n}{n-1}} - 1 = \frac{1}{2n} + O(n^{-2}), \quad s^2 - 1 = \frac{1}{n-1} = O(n^{-1}).$$

Thus, the pointwise global scale modification vanishes asymptotically, and the distinction between the ordinary and surrogate empirical populations is principally a finite-sample covariance-scale effect. The results in Sections 2.5–2.6 characterize why this perturbation does not alter first-order behavior for the smooth functionals satisfying the stated conditions, while the sample-size experiments in Section 5.4 illustrate the decay of the corresponding finite-sample consequences.

The central empirical message is that covariance-scale calibration is exact at the empirical-population level, but its downstream statistical value depends on the response of the relevant functional, tuning rule, and loss to common positive affine rescaling. These guarantees should therefore be distinguished from claims about downstream statistical performance. The construction does not generally improve covariance estimation under Frobenius loss, precision-matrix estimation, individual PCA eigenvalue bias, clustering accuracy, or finite-sample prediction-interval coverage. The numerical studies show why: the map changes global centered scale, but its effect on a downstream procedure depends on the procedure’s objective, regularization, tuning rule, loss, and sensitivity to a common positive affine transformation. In particular, deterministic rescaling can improve or worsen a task-specific finite-sample quantity without constituting a general inferential correction.

The construction is most relevant for compatible continuous feature representations and procedures whose calculations depend on the covariance scale of the empirical support. It can be algebraically equivalent to analytical post-hoc rescaling for covariance-derived quantities with known homogeneity. Its distinct data-layer role arises when a downstream resampling or learning pipeline operates directly on observations and is sensitive to their absolute scale. Conversely, order-based methods, matched nearest-neighbor procedures with consistently transformed query covariates, and matched-initialization K -means can be invariant to the transformation. Procedures that already perform an equivalent normalization or feature-scale adjustment may derive little or no additional benefit.

The asymptotic results are also deliberately limited. The first-order projection result concerns sufficiently smooth population projections, and the bootstrap statement is a conditional transfer principle rather than a standalone bootstrap-validity theorem. Neither result implies nominal finite-sample coverage for percentile or prediction intervals. The undercoverage observed for bootstrap prediction quantiles in Experiment 3 is therefore a boundary-of-validity finding, not a contradiction of the asymptotic analysis.

6.1 Extensions and Limitations

The affine expansion is naturally defined for unconstrained Euclidean feature spaces. It can violate support restrictions for probabilities, proportions, compositions, counts, binary variables, and other bounded or discrete representations. Developing support-preserving analogues requires more than substituting a new loss function: it requires conditions for existence, uniqueness, support preservation, and recovery of appropriate moments or dispersion measures in the relevant geometry.

The method may also be investigated as a preprocessing component for synthetic-data, data-augmentation, or empirical-resampling workflows. It would supply a deterministic covariance-scale modification to the empirical support before a subsequent augmentation or generative procedure. The present results do not establish improvement in synthetic-data fidelity, privacy, diversity, or downstream utility, however, and such claims require dedicated theoretical and empirical study.

Finally, surrogate preprocessing is complementary to changes in resampling design. For example, balanced bootstrap procedures reduce Monte Carlo variation associated with a finite number of resamples, whereas the surrogate map changes the covariance scale of the empirical support before resampling (Srivastav and Srivastav, 2021). These methods target different sources of finite-sample behavior and may be studied jointly. Likewise, conformal prediction targets predictive coverage through a mechanism distinct from covariance-scale calibration; it is a complement for uncertainty quantification, not an alternative implementation of the surrogate map.

6.2 Concluding Remarks

The surrogate population provides an exact representation of the denominator- $(n - 1)$ covariance scale at the level of the empirical support:

$$\hat{\Sigma}_{Y,n} = \frac{n}{n-1} \hat{\Sigma}_{X,n} = S_X.$$

Under the stated coordinate-wise affine restrictions, the map is uniquely characterized; under fixed observation correspondence, it is the unique minimum-displacement configuration satisfying the prescribed mean and covariance constraints.

The method is best interpreted as a data-layer calibration device. For outputs with known homogeneity, it can be algebraically equivalent to post-hoc rescaling. For compatible pipelines that act directly on observations and depend on absolute feature scale, it provides a transparent way to impose the desired empirical-support covariance scale before fitting or resampling. For affine-invariant procedures or constrained feature domains, it can be irrelevant or unsuitable. Accordingly, its principal role is to separate global covariance-scale effects from other finite-sample mechanisms rather than to serve as a universal correction for bootstrap or machine-learning inference.

Complete proofs of Theorems 2, 4, 10, and 11 are given in Online Appendix 1.

Online appendices. Online Appendix 1 contains complete proofs of the affine-uniqueness, minimum-distortion, projection-equivalence, and conditional bootstrap-premise-preservation results. Online Appendix 2 contains additional bootstrap-PCA interval widths, ridge-

sensitivity results, the real-data gene-expression identity illustration, split-conformal reference results, and detailed computational-complexity information.

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Reproducibility. Monte Carlo tables can be regenerated from <https://github.com/asrivas18/covariance-scale-calibration>. The repository records the random-number generator, replication-level seeds, package versions, and exact (n, p, B, M) configurations, together with the scripts that generate the LaTeX tables in the main manuscript and Online Appendix 2. The gene-expression illustration uses the public Bioconductor ALL package (Li, 2009) with a pre-specified probe panel.

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